Lecture (1)

Ch. 7 : Electronic structure of Atoms

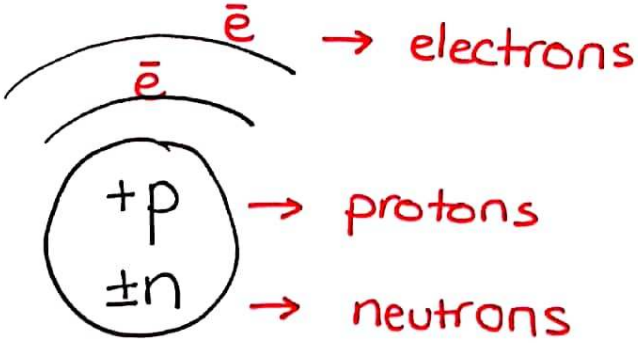
Element : عنصر → Na , Cl, Ca , N , O

Atoms : ذرات

Compound : مركب → NaCl , H_2O , NH_3

- An element is conisists of small particles Called (atoms) ↓
- electronic structure of atoms

- سيتم دراسة التركيب الالكتروني للذرات
- التركيب الالكتروني هو طريقة توزيع الالكترونات حول النواة



nucleas : نواة

- جسيمات سالبة
- جسيمات موجبة
- جسيمات متعادلة

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- كيف تتوزع الالكترونات حول النواة ؟

الالكترون يكون في فلك ، الفلك موجود ضمن مستوى فرعي ، والمستوى الفرعي هو ضمن طبقة رئيسية .

Shells	subshells	orbitals
طبقات رئيسية	طبقات فرعية	افلاك

Quantum numbers : أعداد الكم

"أعداد تصف موقع الالكترون حول النواة"

Shells : Principal Quantum no. → (n)

Subshells : Secondary Quantum no. → (ℓ)

orbitals : magnetic Quantum no. → (m_ℓ)

- $n = 2$ ⇒ $\bar{e}$ found in second shell.
- $\ell = 0$ ⇒ $\bar{e}$ found in subshell "s"

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Quantum numbers :

"n" (Shell)	"ℓ" (Subshell)	"m_ℓ" (Orbital)
	Allowed values :	
↓	↓	↓
1, 2, 3, 4, ... ∞	ℓ = 0, 1, 2, ... (n-1)	-ℓ, .. 0, .. +ℓ.

ℓ	Symbol
0	s
1	p
2	d
3	f

- **example :**

Does an electron with the quantum numbers (n, ℓ, m_ℓ) equal to $(4, 1, 0)$ exist ?

$n = 4$ ✓ $\quad \ell = 1$ ✓ $\quad m_\ell = 0$ ✓

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Which of the following sets of quantum numbers (n, ℓ, m_ℓ) represents an electron in an atom?

a. $(4, 2, 2)$

b. $(6, 5, 3)$

c. $(3, 3, 0)$

d. $(3, -2, 0)$

e. $(1, 0, -1)$

f. $(2, 1, 2)$

g. $(2, 0, 0)$

h. $(4, 2, -3)$

i. $(4, 2, 0)$

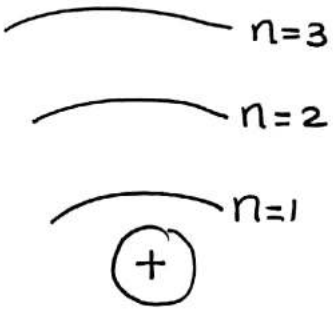
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Quantum Numbers:

1. Principal Quantum no. (n)

- Allowed values : 1, 2, 3, 4, ... ∞

كلما زاد رقم الطبقة للالكترون
زادت طاقته
وزاد حجم الفلك



As "n" increase energy of electron increase, and size of orbital increase.

2. Secondary Quantum no. (ℓ)

- Allowed values : 0, 1, 2, 3, ... (n-1)
  s p d f

Shape of orbital is related to "ℓ"



كروي
"s"

"p"

"d"

"f"

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3. Magnetic Quantum no. (m_ℓ)

Allowed values : $-\ell$ to $+\ell$.

يعتمد على رقم الطبقة الفرعية

S	$\ell = 0$	$m_\ell = 0$.
p	$\ell = 1$	$m_\ell = -1, 0, 1$
d	$\ell = 2$	$m_\ell = -2, -1, 0, 1, 2$
f	$\ell = 3$	$m_\ell = -3, -2, -1, 0, 1, 2, 3$

- How many orbitals are found in :

S - subshell $\Rightarrow$ 1 orbital ($m_\ell = 0$)

p- subshell $\Rightarrow$ 3 orbitals ($m_\ell = -1, 0, 1$)

d- subshell $\Rightarrow$ 5 orbitals ($m_\ell = -2, -1, 0, 1, 2$)

f- subshell $\Rightarrow$ 7 orbitals ($m_\ell = -3, -2, -1, 0, 1, 2, 3$)

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1 أفلك	s	$\frac{}{0}$	$\ell = 0$	$m_\ell = 0$
3 أفلك	p	$\frac{}{-1}\ \frac{}{0}\ \frac{}{1}$	$\ell = 1$	$m_\ell = -1, 0, 1$
5 أفلك	d	$\frac{}{-2}\ \frac{}{-1}\ \frac{}{0}\ \frac{}{1}\ \frac{}{2}$	$\ell = 2$	$m_\ell = -2, -1, 0, 1, 2$
7 أفلك	f	$\frac{}{-3}\ \frac{}{-2}\ \frac{}{-1}\ \frac{}{0}\ \frac{}{1}\ \frac{}{2}\ \frac{}{3}$	$\ell = 3$	$m_\ell = -3, -2, -1, 0, 1, 2, 3$

(4) Spin Quantum no. (m_s)

Allowed values : $+\frac{1}{2}$ or $-\frac{1}{2}$

العدد الكمي المغزلي له علاقة بدوران الإلكترون حول محوره "إما مع عقارب الساعة أو عكس عقارب الساعة"



$+\frac{1}{2}$ $\quad$ $-\frac{1}{2}$

$-\frac{1}{2}$ $\quad$ $+\frac{1}{2}$

Clockwise $\quad$ Counterclockwise.

m_s : $+\frac{1}{2}$ or $-\frac{1}{2}$

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Summary :

n	ℓ	Name	m_ℓ.	no. of orbitals
1	0	1s	0	(1)
2	0	2s	0	(4)
	1	2p	-1 0 1	
3	0	3s	0	(9)
	1	3p	-1 0 1	
	2	3d	-2 -1 0 1 2	
4	0	4s	0	(16)
	1	4p	-1 0 1	
	2	4d	-2 -1 0 1 2	
	3	4f	-3 -2 -1 0 1 2 3	

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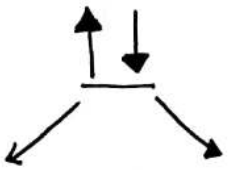
Pauli° exclusion principle

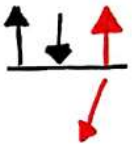
"مبدأ باولي للإستبعاد."

لا يمكن لإلكترونين في نفس الذرة أن يتشابها
في أعدادهما الكمية الأربعة

وعليه الفلك الواحد يتسع لإلكترونين كحد أقصى.

ex) 3s





$n=3$	$n=3$	$n=3$
$\ell=0$	$\ell=0$	$\ell=0$
$m_\ell=0$	$m_\ell=0$	$m_\ell=0$
$m_s=+1/2$	$m_s=-1/2$	$m_s=+1/2$

S — 2ē as a maximum

p ——— 6ē as a maximum

d ————— 10 ē as a maximum

f ——————— 14 ē (maximum)

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n=1	1s	1 subshell
n=2	2s , 2p	2 subshells
n=3	3s, 3p, 3d	3 subshells.
n=4	4s , 4p, 4d , 4f	4 Subshells
n=5	5s , 5p, 5d , 5f, 5g.	5 subshell

How many subshells are found in 4th shell?

4.

How many orbitals are found in the fourth shell (n=4) = 16

Calculate maximum no. of $\bar{e}$s that could be occupied in the fourth shell.

4s	4p	4d	4f	
↓	↓	↓	↓	
$2\bar{e}$	$6\bar{e}$	$10\bar{e}$	$14\bar{e}$	= $32\bar{e}$

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Hunds Rule:

Electrons are distributed among degenerate orbitals in a way that gives maximum no. of unpaired ēs.

4d _ _ _ _ _ five degenerate orbitals.

أفلاك متساوية في الطاقة

$4d^7$ ⇅ ⇅ ↑ ↑ ↑ ⇒ Stable

3 unpaired ēs.

$4d^7$ ⇅ ⇅ ⇅ ↑ _ ⇒ unstable.

↳ one unpaired ē

قاعدة هوند : توزّع الالكترونات على الأفلاك المتساوية في الطاقة فرادى بنفس إتجاه العزل، ثم يحدث إزدواج للإلكترونات.

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Examples of Hund's Rule:

$6p^2$

$3d^5$

$6d^8$

$2s^2$

$4f^6$

$3f^{11}$ X

$5p^8$ X

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Ground State Configuration

1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p

6s 4f 5d 6p 7s 5f 6d 7p

Energy of orbitals increase

عند توزيع الإلكترونات يراعى هذا الترتيب في التوزيع

Aufbau principle:

قانون البناء الذري

والذي نحصل من خلاله على التوزيع الأكثر إستقرار للذرة

Ground State configuration (G.S) ← التوزيع الالكتروني الاكثر استقراراً.

① نراعي ترتيب الطاقة عند توزيع الالكترونات

② نراعي قاعدة هوند (عند توزيع الالكترونات على الافلاك المتساويه في الطاقة)

③ نراعي مبدأ باولي

كل فلك يتسع لالكترونين في الفلك (باتجاه دوران متعاكس)

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Write down the ground state configuration of the following elements

1s 2s 2p 3s 3p 4s 3d 4p

5s 4d 5p 6s 4f 5d 6p

$_{11}Na$ (atomic no.)

$_{11}Na$: $1s^2\ 2s^2\ 2p^6\ 3s^1$

↑↓ ↑↓ ↑↑↑ ↑ one unpaired $\bar{e}$

$_{7}N$: $1s^2\ 2s^2\ 2p^3$

↑↓ ↑↓ ↑ ↑ ↑ G.S

↑↓ ↑↓ ↓ ↓ ↓ G.S

three unpaired $\bar{e}_s$.

$1s^2\ 2s^2\ 2p^2\ 3s^1$ Not G.S

$1s^2\ 2s^2\ 2p^3$

↑↑ ↑↓ ↑ ↓ ↑ Not G.S

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Ground State configurations in terms of Noble gases.

- Noble elements: are stable unreactive elements.

عناصر خاملة مستقرة
لاتتفاعل نظراً
لاستقرارها العالي

$_{2}He : 1s^2$

$_{10}Ne : 1s^2 2s^2 2p^6$

$_{18}Ar : 1s^2 2s^2 2p^6 3s^2 3p^6$

$_{36}Kr : 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$

$_{54}Xe :$

$_{86}Rn :$

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نستطيع كتابة التوزيعات الالكترونية للعناصر بطلقة العناصر النبيلة .

Examples:

$_{6}C$: $1s^2 2s^2 2p^2$: $[He]\, 2s^2 2p^2$

$_{16}S$ $1s^2 2s^2 2p^6 3s^2 3p^4$: $[Ne]\, 3s^2 3p^4$

$_{20}Ca$: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$: $[Ar]\, 4s^2$

$_{32}Ge$: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$

$[Ar]\, 4s^2 3d^{10} 4p^2$

$_{38}Sr$: $[Kr]\, 5s^2$

Noble : $_{2}He$ ، $_{10}Ne$ ، $_{18}Ar$ ، $_{36}Kr$ ، $_{54}Xe$

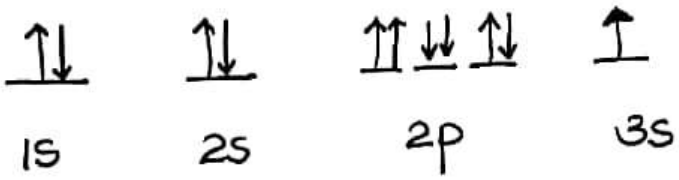
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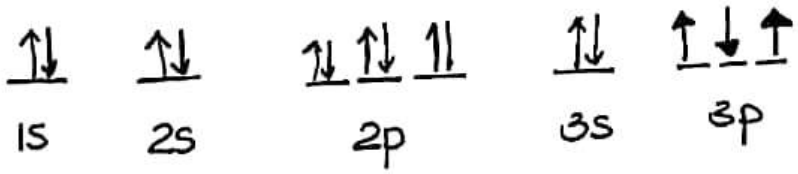
Which of the following electronic configurations is a ground state configuration

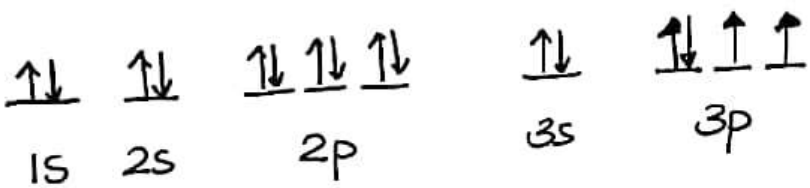
a. $1s^2 2s^2 2p^2 3s^2$

b. $[Ne]\, 4s^2$

c. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5 4p^3$

d. 

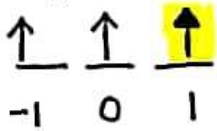
e. 

f. 

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Write down the set of quantum numbers for the last $\bar{e}$ in: "أيضاً حدد عدد الالكترونات المنفردة (unpaired $\bar{e}_s$)

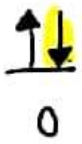
$_9F$: $1s^2 2s^2 2p^3$ ⇒ 3 unpaired $\bar{e}$s.



(n, ℓ, m_ℓ, m_s)

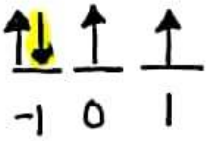
$(2, 1, 1, \frac{1}{2})$

$_{20}Ca$ $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ ⇒ 0 unpaired $\bar{e}$



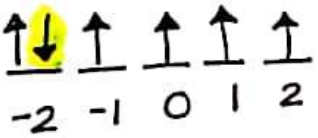
$(4, 0, 0, \frac{1}{2})$

$_{34}Se$ $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^4$ ⇒ 2 unpaired $\bar{e}$



$(4, 1, -1, \frac{1}{2})$

$_{26}Fe$ $[Ar]\ 4s^2 3d^6$ 4unpaired $\bar{e}$



$(3, 2, -2, \frac{1}{2})$

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For the element $_{25}Mn$

$1s^2$ $2s^2$ $2p^6$ $3s^2$ $3p^6$ $4s^2$ $3d^5$

↑↓ ↑↓ ↑↓↑↓↑↓ ↑↓ ↑↓↑↓↑↓ ↑↓ ↑ ↑ ↑ ↑ ↑

1) How many electrons have $\ell = 1$?

"p" electrons = 6 + 6 = 12

2) How many electrons have $n = 3$?

$3s^2\ 3p^6\ 3d^5$ = 2+6+5 = 13 $\bar{e}$

3) How many electrons have $m_\ell = 0$?

2+ 2 + 2 +2 +2 +2 +1 = 13 $\bar{e}$

4) How many $\bar{e}$s have $m_s = ½$, $-½$?

$(½)$ ↑ ⇒ 15 $\bar{e}_s$

$(-½)$ ↓ ⇒ 10 $\bar{e}_s$

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Types of elements in periodic table :

أنواع العناصر في الجدول الدوري :

1) Representative elements العناصر الممثلة

"عناصر ينتهي توزيعها الالكتروني بـ "s" أو "p"

ex) $_{7}N$: $1s^2 2s^2 2p^3$

$_{11}Na$: $1s^2 2s^2 2p^6 3s^1$

$_{32}Ge$: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$

2) Transtion elements: العناصر الانتقاليه

"عناصر ينتهي توزيعها الالكتروني بـ "d"

ex) $_{27}Co$: $[Ar]\ 4s^2 3d^7$

$_{41}Nb$: $[Kr]\ 5s^2 4d^3$

3) Inner transtion element : عناصر انتقاليه داخليه

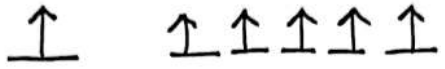
ينتهي توزيعها بـ "f"

ex) $_{62}Sm$: $[Xe]\ 6s^2 4f^6$

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Exceptions in electronic configurations for "Transtion Elements"

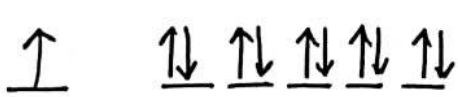
$_{24}Cr$: $[Ar]\ 4s^1\ 3d^5$ (G.S)



Transtion elements $d^4 \rightarrow d^5$

$d^9 \rightarrow d^{10}$

$_{29}Cu$ $[Ar]\ 4s^1\ 3d^{10}$



هذه الاستثناءات في التوزيع الالكتروني المستقر
هي للعناصر الانتقالية
أما العناصر الممثلة فلا يوجد فيها استثناءات

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Magnetic Properties of Elements.

(1) Paramagnetic elements:

- elements that have one or more unpaired $\bar{e}$s
- They are stongly attracted toward a magnet.

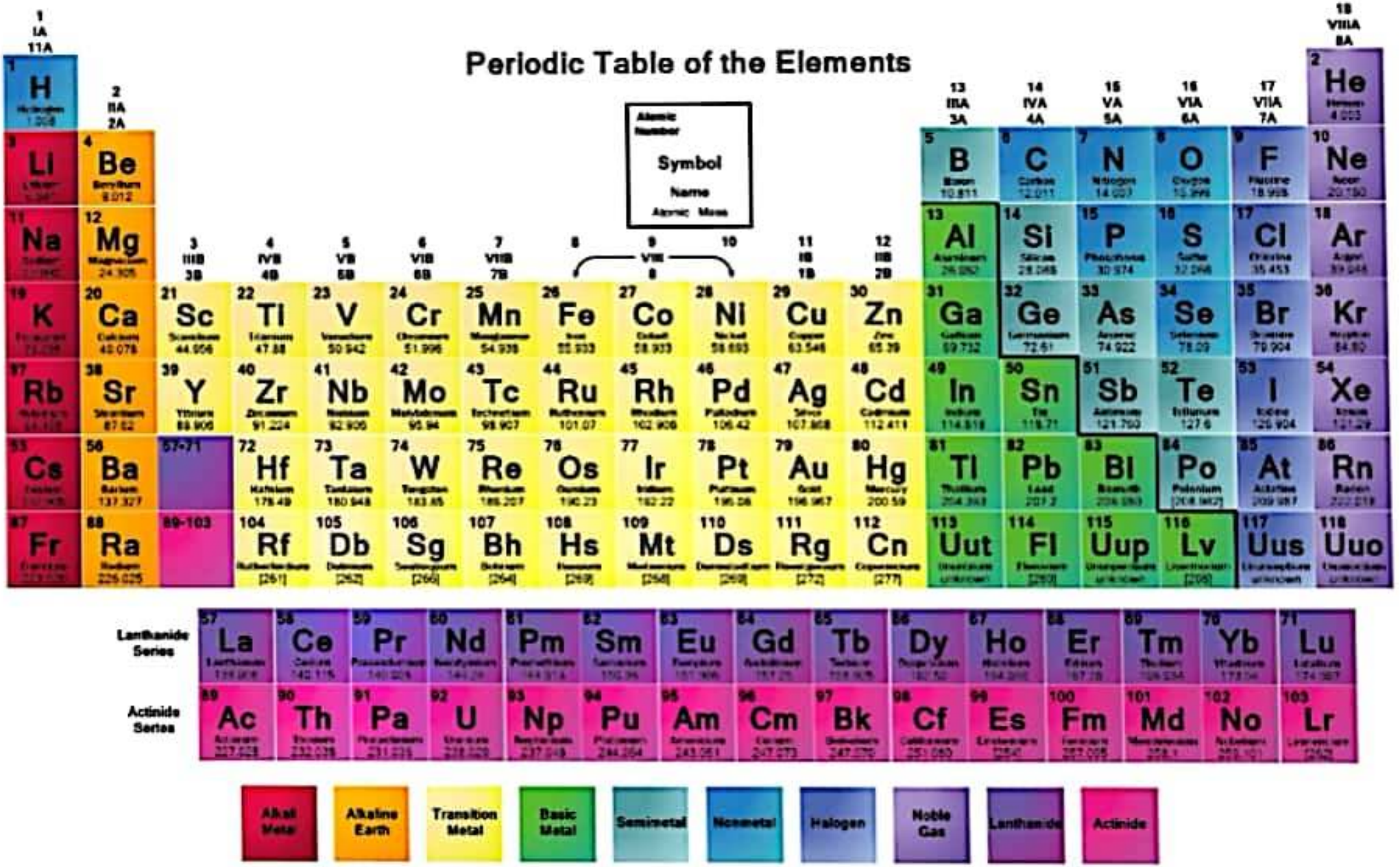
ex) $_{7}N$, $_{6}C$, $_{24}Cr$

(2) Diamagnetic Elements:

- Elements that do not have unpaired electrons (zero unpaired $\bar{e}$s).
- These elements are weakly attracted toward a magnet.

ex) $_{18}Ar$, $_{12}Mg$, $_{30}Zn$

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Ch. 8

Periodic Table: الجدول الدوري

- Elements in periodic table:

1) Representative elements → s- block, p- block

2) Transtion Elements d- block

3) Inner transtion elements f- block

- Families → Family "A" ⇒ Representative elements.
  Family "B" ⇒ Transtion and Inner transtion elements.

- Columns ⇒ Groups : أعمدة ، مجموعات
- Rows ⇒ periods أسطر ، دورات

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(1) Representative elements.

(S-block) and (p-block)

Groups 1A → 8A

1A : Alkali metals. الفلزات القلوية

2A : Alkaline earth metals. الفلزات القلويه الترابية

7A: Halogens الهالوجينات

8A: Noble elements. العناصر النبيلة .

Valence shell : الغلاف الخارجي ، المستوى الأخير

(ns) ex) $_{12}Mg : 1s^2 2s^2 2p^6 \underline{3s^2}$

Valence shell : $3s^2$

(nsnp) ex) $_{8}O : 1s^2 \underline{2s^2 2p^4}$

Valence shell : $2s^2 2p^4$

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للعناصر الممثلة :

- رقم المستوى الأخير (Valence Shell) يمثل رقم الدورة الموجود فيها العنصر
- عدد الالكترونات في المستوى الخارجي يمثل رقم المجموعة

ex) $_{34}Se : 1S^2\ 2S^2 2p^6\ 3S^2 3p^6\ 4S^2 3d^{10}\ 4p^4$

Valence shell : $4S^{(2)} 4p^{(4)}$ ⇒ summation = group no.

رقم المجموعة

رقم الدورة : period no.

period no. = 4

موجود في السطر الرابع في الجدول الدوري

group no. = $2 + 4\bar{e} = 6\bar{e}$

group 6A

موجود في المجموعة السادسة (A)

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- 1A (IA): Alkali metals
- 2A (IIA): Alkaline earth metals
- 7A (VIIA): Halogens.
- 8A (VIIIA): Noble elements

- Representative elements (family A)
- Transtion and inner transtion elements (family B)

Rep. (S-block)

Representative elements (p-block)

(Periodic Table)

period	1A	2A	Transtion elements (d-block)										3A	4A	5A	6A	7A	8A
			3B	4B	5B	6B	7B	8B			1B	2B						
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg											13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	57 La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	89 Ac	104 Rf	105 Db	106 Sg	107 Bh	108 Hs	109 Mt	110	111	112	113	114				

58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu	→ Lanthanides (4f)
90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr	→ Actinides (5f)

Inner transtion elements (f-block)

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Representative elements (metallic Character)

من حيث الخواص الفلزية

(1) Metals. فلزات

(2) Metalloids (Semimetals) أشباه فلزات

(3) Nonmetals لافلزات

(2) Transition elements (d-block)

Family "B"

Valence shell ns (n-1)d

ex) $_{27}Co$: $[Ar]\,4s^2\,3d^7$

Valence shell

period no. = 4

للعناصر الانتقالية رقم أعلى (s) هو يمثل رقم الدورة .

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(3) Inner Transtion Elements

(f- block)

period (6) ⇒ Lanthanides. اللانثانيدات

period (7) ⇒ Actinides الآكتينيدات

توزيعهم الإلكتروني فيه الكثير من الاستثناءات لكن
(غير مطلوبة)

Examples:

(1) Give an example of a nonmetal element.

(2) Give an example of a Halogen.

(3) Give an example of an actinide element.

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(4) Give an example of an element that have one electron in the fourth shell.

(5) Give an example of a transtion element that have five unpaired $\bar{e}_s$ in its valence shell.

(6) What is the element which belongs to period (5) group (4B)

(7) Give an example of an element that is paramagnetic.

(8) Give an example of a diamagnetic transtion element.

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Periodic Properties of Representative elements

سنـدرس الخواص الدورية للعـناصر الممثلة (العائلة A)

(1) Atomic size — الحجم الذري

(2) Ionization Energy — طاقة التأين

(3) Electron Affinity — الألفة الالكترونية

(4) Metallic Character. — الخواص الفلزية

(1) Atomic Size: الحجم الذري

A : $1s^2 2s^1$ — 2 shells.

B : $1s^2 2s^2 2p^6 3s^1$ — 3 shells

C : $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ — 4 shells.

$C > B > A$: atomic size

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• عندما يزداد عدد المدارات للذرات ، فإن حجمها الذري يزداد .

$_{9}F$	$_{11}Na$	$_{20}Ca$
$1s^2 2s^2 2p^5$	$1s^2 2s^2 2p^6 3s^1$	$[Ar] 4s^2$
$n=2$	$n=3$	$n=4$

$$Ca > Na > F$$

• لكن عندما يتساوى عدد المدارات لعدة ذرات فإنه العامل الذي سيتحكم بالحجم هو شحنة النواة الفعالة

Effective nuclear Charge (Z^*)

• وهي مقدار الجذب الفعلي بين النواة الموجبة والكترونات الغلاف الخارجي

أو مقدار ما تشعر به الكترونات الغلاف الأخير من النواة الموجبة (كونه يوجد الكترونات تحجب شحنة النواة الموجبة عن الكترونات المدار الأخير) .

* كلما زادت (Z^*) قل حجم الذرة .

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Arrange the following atoms according to decrease in their atomic sizes.

$_{16}S : 1s^2 2s^2 2p^6 3s^2 3p^4$. $Z^* = 16 - 10 = +6$

$_{11}Na : 1s^2 2s^2 2p^6 3s^1$ $Z^* = 11 - 10 = +1$

$_{15}P : 1s^2 2s^2 2p^6 3s^2 3p^3$ $Z^* = 15 - 10 = +5$

$_{11}Na > _{15}P > _{16}S$

+1 +5 +6.

علاقة عكسية بين (Z^*) والحجم الذري .

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How to compare atoms according to their atomic size.

① ترتيب العناصر حسب عدد الطبقات

(علاقة طردية)

② إذا تساوى عدد الطبقات ننظر إلى

الحد الذري للعنصر (علاقة عكسية)

الحد الذري يعكس شحنة النواة الفعالة.

Scanned with CamScanner

Arrange the following atoms according to decrease in their atomic size.

$_{17}A$ $_{14}B$ $_{32}C$ $_{12}D$ $_{20}E$ $_{38}F$

A : $[Ne]\ 3s^2 3p^5$

B : $[Ne]\ 3s^2 3p^2$

C : $[Ar]\ 4s^2 3d^{10} 4p^2$

D : $[Ne]\ 3s^2$

E : $[Ar]\ 4s^2$

F : $[Kr]\ 5s^2$

F > E > C > D > B > A

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Atomic Size in Periodic table.

(Group and Period)

1A	2A	3A	4A	5A	6A	7A	8A.	
$_{3}Li$	$_{4}Be$	$_{5}B$	$_{6}C$	$_{7}N$	$_{8}O$	$_{9}F$	$_{10}Ne$	⇒ 2 shells
$_{11}Na$								⇒ 3 shells
$_{19}K$								⇒ 4 shells
$_{37}Rb$								⇒ 5 shells
$_{55}Cs$								⇒ 6 shells
$_{87}Fr$								⇒ 7 shells.

In a group

Top

↓

bottom

- no. of shells increase
- Size increase.

In a period.

Left → right

- no. of shells is constant
- Z^* increase
- Size decrease

Scanned with CamScanner

Periodic Properties of Representative elements:

(2) Ionization energy (IE)

طاقة تأين العنصر هي الطاقة اللازمة لنزع إلكترون الأضعف إرتباطاً (الأبعد عن النواة) من الذرة المتعادلة في حالتها الغازية.

$$X_{(g)} \longrightarrow X^{+}_{(g)} + \bar{e}$$

$$_{8}O \longrightarrow _{8}O^{+} + \bar{e}$$

$1S^2 2S^2 2p^4$ $\qquad$ $1S^2 2S^2 2p^3$

+8 protons $\qquad$ (+8) protons

8 electrons. $\qquad$ 7 electrons

IE_1 for oxygen = 1314 KJ/mol.

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First ionization energy for representative elements, is affected by these factors.

(1) No. of shells.

(2) Effective nuclear charge. (Z^*)
for elements in same period

للعناصر التي لديها نفس عدد الطبقات (موجودين في نفس الدورة في الجدول الدوري) نرتبهم حسب العدد الذري (علاقة عكسية) لكن يوجد بعض الاستثناءات المفروض :

8A > 7A > 6A > 5A) > 4A > 3A > 2A > 1A.

لكن في الواقع

8A > 7A > 5A > 6A > 4A > 2A > 3A > 1A

لعناصر الدورة الواحدة .

Scanned with CamScanner

$_{3}Li$ $1s^2\underline{2s^1}$ الإلكترون الأخير أقرب للنواة

$_{11}Na$ $1s^2 2s^2 2p^6 \underline{3s^1}$

$_{19}K$ $[Ar]\ \underline{4s^1}$ الإلكترون الأخير أبعد عن النواة.

$Li > Na > K.$ (IE_1)

Generally : in a group going from top to bottom, IE_1 will decrease (because no. of shells increase).

Scanned with CamScanner

Through a period:

1A	2A	3A	4A	5A	6A	7A	8A
Li	Be	B	C	N	O	F	Ne
3	4	5	6	7	8	9	10
520	899	801	1086	1420	1313	1681	2080

higher than expected.

2A > 3A.

higher than expected.

5A > 6A.

$Be: 1s^2 2s^2$

$Be^+: 1s^2 2s^1$ (توزيع أقل استقرار)

$N: 1s^2 2s^2 2p^3$

$N^+: 1s^2 2s^2 2p^2$

كـ

توزيع أقل استقرار مقارنة بتوزيع النيتروجين

قبل إزالة الالكترون

Scanned with CamScanner

Arrange the following elements according to their atomic sizes and their IE_1.

$_{15}A$	$_{7}B$	$_{35}C$	$_{16}D$	$_{6}E$	$_{11}F$	$_{10}G$
↓	↓	↓	↓	↓	↓	↓
n=3	n=2	n=4	n=3	n=2	n=3	n=2 (noble)
↓	↓	↓	↓	↓	↓	↓
5A	5A	7A	6A	4A	1A	8A.

Atomic size:

C > F > A > D > E > B > G.

Ionization energy:

G > B, E > F, A, D > C

(B, E: ترتب حسب رقم المجموعة — F, A, D: ترتب حسب رقم المجموعة)

B > E A > D > F

G > B > E > A > D > F > C

Scanned with CamScanner

(3) Electron Affinity (EA) الألفة الإلكترونية

$$X_{(g)} + \bar{e} \longrightarrow X^-_{(g)}$$

Energy associated with addition of an electron to a neutral atom in its gaseous state.

In general :

(1) Through a group going from top to bottom, no. of shells increase and EA decrease.

(2) Through a period : going from left to right, Z^* increase, and EA increase.

(3) Noble elements have the lowest EA compared to elements in same period.

Scanned with CamScanner

(4) Halogens have the highest EA compared to elements in the same period.

Example:

$_{10}A$ $_{4}B$ $_{9}C$ $_{6}D$.

A: $1s^2 2s^2 2p^6$ (Noble)

B: $1s^2 2s^2$

C: $1s^2 2s^2 2p^5$ (7A, halogen)

D: $1s^2 2s^2 2p^2$

C > D > B > A

Scanned with CamScanner

(4) Metallic Character:

"الخواص الفلزية للعناصر"

In a group: from top to bottom metallic properties increase.

In a period: from left to right metallic properties decrease.

Scanned with CamScanner

Ionic sizes:

Ions → cations أيونات موجبة

Ions → anions أيونات سالبة

- الأيون الموجب هو ذرة متعادلة فقدت إلكترون أو أكثر.
- الأيون السالب هو ذرة متعادلة اكتسبت الكترون أو أكثر.

$_{11}Na$

$1s^2 2s^2 2p 3s^1$

3 shells

$_{11}Na^+$

$1s^2 2s^2 2p^6$.

2 shells.

$Na > Na^+$

$_{6}C$

$1s^2 2s^2 2p^2$

2 shells

$_{6}C^+$

$1s^2 2s^2 2p^1$

2shells

$C > C^+$

Scanned with CamScanner

- All neutral atoms are larger in size than their cations.

- As positive charge of ion increase size decrease.

$$Fe > Fe^{2+} > Fe^{3+}$$

- On the otherside, all neutral atoms are smaller in size than their anions.

$$Cl < Cl^{-}$$

- As negative sign increase, size decrease.

$$O < O^{-} < O^{2-}$$

Scanned with CamScanner

How many unpaired electrons does $_{26}Fe^{3+}$ have ?

$_{26}Fe$: $[Ar]\ 4s^2\ 3d^6$

Fe^{3+}: $[Ar]\ 3d^5$ ↓ ↓ ↓ ↓ ↓ (5)

Transtion elements lose $\bar{e}_s$ ↑ from the "s" subshell, before losing $\bar{e}_s$ from "d" subshell.

Scanned with CamScanner

Stoichiometery

$$\text{moles} = \frac{\text{mass}}{\text{At.wt}} = \frac{\text{mass}}{\text{Molar mass}}$$

At.wt: للعناصر

Molar mass: للمركبات

$$\text{moles} = \frac{\text{no. of (atoms, molecules, ions)}}{6.022 \times 10^{23}}$$

Chemical formula: الصيغة الكيميائية

A_2B_3.

1mol A_2B_3 → 2mol "A"

1mol A_2B_3 → 3 mol "B"

Scanned with CamScanner

- Calculate number of "Na" atoms found in 250 g of Na.

Na = 23 g/mol

6.022×10^{23}

$$\text{moles "Na"} = \frac{\text{mass "Na"}}{\text{At.wt (Na)}}$$

$$= \frac{250\ g}{23\ g/mol} = 10.86\ mol\ Na.$$

$$\text{number of "Na" atoms} = 10.86\ mol \times 6.022 \times 10^{23}$$

$$= 6.54 \times 10^{24}.$$

$$\text{mass} = \text{moles} \times \text{At.wt}$$

$$\text{number} = \text{moles} \times 6.022 \times 10^{23}$$

Scanned with CamScanner

Calculate mass of 10 atoms of "Na"

Na = 23g/mol

$$\text{moles "Na"} = \frac{\text{number}}{6.022 \times 10^{23}} = \frac{10 \text{ atoms}}{6.022 \times 10^{23}}$$

$$= 1.66 \times 10^{-23} \text{ mol "Na"}$$

$$\text{mass "Na"} = \text{moles} \times \text{At.wt}$$

$$= 1.66 \times 10^{-23} \times 23 \text{ g/mol}$$

$$= 3.82 \times 10^{-22} \text{ gram.}$$

Scanned with CamScanner

How many molecules of H_2O are found in 100 g of water.

$H = 1$ g/mol $O = 16$ g/mol

$H_2O = 18$ g/mol.

$$\text{moles } "H_2O" = \frac{100\text{g}}{18\text{g/mol}} = 5.56 \text{ mol } H_2O$$

$$\text{number of } "H_2O" \text{ molecules} = \text{moles} \times (6.022 \times 10^{23})$$

$$= 5.56 \times 6.022 \times 10^{23}$$

$$= 3.34 \times 10^{24} \text{ molecule.}$$

العناصر مكونة من ذرات (atoms)

المركبات مكونة من جزيئات (molecules)

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Al_2S_3

Al = 27 g/mol

S = 32 g/mol

How many grams of "S" are required to completely combine with 15g of "Al" in Al_2S_3 ?

2 mol "Al" $\Leftrightarrow$ 3 mol "S"

0.56 mol $\left(\frac{15\ g}{27\ g/mol}\right) \Leftrightarrow$?

moles of "S" $= \frac{0.56 \times 3}{2}$

$= 0.84$ mol "S"

mass "S" $= 0.84$ mol $\times 32$ g/mol

$= 26.88$ gram "S"

Scanned with CamScanner

ex) How many "H" atoms are found in 1.67 g of C_3H_7OH ?

C = 12 H = 1 O = 16 g/mol

C_3H_7OH = 60 g/mol

1.67 g C_3H_7OH $\longrightarrow$ moles C_3H_7OH

$\div$ M.wt

$\downarrow$ 1 : 8 استخدام الصيغة

no. of "H" atoms $\longleftarrow$ moles "H"

$\times 6.022 \times 10^{23}$

$$\text{moles } C_3H_7OH = \frac{1.67\text{ g}}{60\text{ g/mol}} = 0.028 \text{ mol } C_3H_7OH$$

1 mol C_3H_7OH $\Leftrightarrow$ 8 mol "H"

0.028 $\Leftrightarrow$?

$$\text{moles "H"} = \frac{0.028 \times 8}{1} = 0.224 \text{ mol "H"}$$

Scanned with CamScanner

No. of "H" atoms = moles $\times 6.022 \times 10^{23}$

$= 0.224 \text{ mol} \times 6.022 \times 10^{23}$

$= 1.349 \times 10^{23}$ "H" atom.

mass "X" $\xrightarrow[\div \text{M.wt.}]{\div \text{At.wt}}$ moles "X"

no. of "X" atoms $\xrightarrow{\div 6.022 \times 10^{23}}$ moles of "X"

moles "X" $\xrightarrow[\text{formula}]{\text{Chemical}}$ moles of "Y"

(نسبة وتناسب)

Scanned with CamScanner

Find mass of the compound $Ca_3(PO_4)_2$ that contains 18 gram of Oxygen :

Ca = 40

P = 31 g/mol

O = 16 g/mol

$Ca_3(PO_4)_2$ = 310 g/mol

18 g "O" $\xrightarrow{\div 16}$ moles "O"

$\downarrow$ 1 $Ca_3(PO_4)_2$: 8 "O"

? :

mass $Ca_3(PO_4)_2$ $\xleftarrow{\times 310.}$ moles $Ca_3(PO_4)_2$

$$\text{mass } Ca_3(PO_4)_2 = \frac{18}{16} \times \frac{1}{8} \times 310$$

$$= 43.6 \text{ gram.}$$

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Ch.3 Stoichiometery

Percentage Composition:

$$\% \text{ element} = \frac{\text{mass of element}}{\text{mass of whole compound}} \times 100\%$$

$$\% \text{ element} = \frac{(\text{At.wt} \times \text{subscript})}{\text{Molar mass}} \times 100\%$$

ex) For P_2O_5

$P = 31$, $O = 16$, $P_2O_5 = 142$ g/mol

$$\% P = \frac{31 \times 2}{142} \times 100 = 43.66\%$$

$$\% \text{ "O"} = \frac{16 \times 5}{142} \times 100 = 56.34\%$$

Scanned with CamScanner

Calculate mass percentages of C, H, and Oxygen in 70 gram $C_2H_2O_4$

C = 12 , H = 1 , O = 16 g/mol.

$C_2H_2O_4$ = 90 g/mol.

$$\%C = \frac{2 \times 12}{90} \times 100\% = 26.7\%$$

$$\%H = \frac{2 \times 1}{90} \times 100\% = 2.2\%$$

$$\%\text{"O"} = \frac{4 \times 16}{90} \times 100\% = 71.1\%$$

mass of "C" = 18.69 g.

mass of "H" = 1.54 g.

mass of "O" = 49.77 g

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Molecular formula: تعبر عن العدد الحقيقي للذرات في المركب

Empirical formula: تعبر عن أبسط نسبة أعداد صحيحة بين ذرات المركب

C_6H_6 

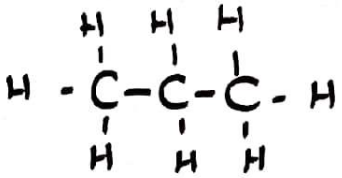
C : H

1 : 1

(CH)

الصيغة الأولية

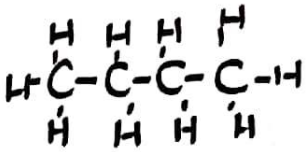
الصيغة الجزيئية

C_3H_6 

C : H

3 : 6

1 : 2 $\Rightarrow (CH_2)$

C_4H_8 

C : H

4 : 8

1 : 2 $\Rightarrow (CH_2)$

$$\text{Molecular formula} = (\text{emp. formula})_n$$

n: integer. عدد صحيح

$$n = \frac{\text{Molar mass of compound.}}{\text{Molar mass of emp. formula.}}$$

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ex) Emp. formula : CH_2 (14 g/mol)

Molecular formula ?

$(CH_2)_n$ = molecular formula.

$$n = \frac{56\,g/mol}{14\,g/mol} = 4$$

Molecular formula = $(CH_2)_4 = C_4H_8$.

Molecular	Empirical
H_3PO_4	H_3PO_4
H_2O	H_2O
Al_2O_3	Al_2O_3
$C_6H_{12}O_6$	CH_2O
C_2H_4	CH_2
C_3H_6	CH_2
C_6H_{12}	CH_2
C_6H_6	CH

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- كيف نجد الصيغة الأولية لمركب معين :

① إيجاد مولات كل عنصر في المركب

" تحويل الكتل ، العدد ، النسب إلى مولات "

② إيجاد أبسط نسبة أعداد صحيحة بين الذرات في المركب

" بقسمة على أصغر عدد ثم التقريب وأحياناً ضرب الصيغة كاملة " بعدد صحيح .

③ لإيجاد الصيغة الجزيئية "الحقيقية "

يجب إيجاد العدد الصحيح (n)

نضرب (n) بالصيغة الأولية لإيجاد الصيغة الحقيقية " الجزيئية "

Scanned with CamScanner

H	C	O
3.22 %	19.35 %	77.43 %

C = 12
H = 1
O = 16

M.wt of compound = 62 g/mol.

what is its molecular formula.

H	C	O
3.22 g	19.35 g	77.43 g.
↓	↓	↓
3.22 mol "H"	1.61 mol "C"	4.84 mol "O"
↓	لتحويلها إلى أعداد صحيحة ↓	↓
2	1	3

H_2CO_3 : emp. formula. $\Rightarrow$ 62 g/mol

$$n = \frac{62\ g/mol}{62\ g/mol} = 1 \Rightarrow H_2CO_3$$

molecular formula

Scanned with CamScanner

IF molar mass = 222 g/mol

$$n = \frac{222 \text{ g/mol}}{74 \text{ g/mol}} = 3$$

$$(C_3H_{10}N_2)_3 = C_9H_{30}N_6$$

molecular formula.

After dividing by smallest no. of moles to get whole no. ratio.

Round ⇒ □.1 , □.2 , □.8 , □.9

But for □.3 (↓x3) □.4 (↓x2) □.5 (↓x2) □.6 (↓x2) □.7 (↓x3)

ضرب الصيغة كاملة"

Scanned with CamScanner

Find the molecular formula for a compound that contains 48.7 % of "C" 8.1 % of "H" and 43.2 % of "O".

If molecular weight of compound = 226

Given that:

C = 12 , O = 16 , H = 1 g/mol.

Scanned with CamScanner

H	C	O
3.22 %	19.35 %	77.43 %

C = 12
H = 1
O = 16

M.wt of compound = 62 g/mol.

what is its molecular formula.

H	C	O
3.22 g	19.35 g	77.43 g.
↓	↓	↓
3.22 mol	1.61 mol	4.84 mol
H	"C"	"O"
↓	لتحويلها إلى أعداد صحيحة	↓
	↓	
2	1	3

H_2CO_3 : emp. formula. $\Rightarrow$ 62 g/mol

$$n = \frac{62\ g/mol}{62\ g/mol} = 1 \Longrightarrow H_2CO_3$$

molecular formula

Scanned with CamScanner

ex) 54.26g compound (C, H, N)

C	H	N
25.2g	7.5g	21.56g
↓	↓	↓
moles "C"	mols "H"	moles "N"
2.1 mol	7.5 mol	1.54 mol.
↓ ÷1.54	↓ ÷1.54	↓ ÷1.54
1.4	4.9	1
لاتقرب	5 نقرب	

Multiply by smallest integer to remove .4

$$(C_{1.4}\ H_5\ N)_2 = C_{2.8}\ H_{10}\ N_2$$

↓

$$C_3\ H_{10}\ N_2$$

emp. formula = 74 g/mol

Scanned with CamScanner

If molar mass = 222 g/mol

$$n = \frac{222\ g/mol}{74\ g/mol} = 3$$

$$(C_3H_{10}N_2)_3 = C_9H_{30}N_6$$

molecular formula.

After dividing by smallest no. of moles to get whole no. ratio.

Round ⇒ □.1 , □.2 , □.8 , □.9

But for □.3 □.4 □.5 □.6 □.7

↓x3 ↓x2 ↓x2 ↓x2 ↓x3

ضرب الصيغة كاملة"

Scanned with CamScanner

Find the molecular formula for a compound that contains 48.7 % of "C" 8.1 % of "H" and 43.2 % of "O".

If molecular weight of compound = 226

Given that:

C = 12 , O = 16 , H = 1 g/mol.

Scanned with CamScanner

C	H	O
48.7g	8.1 g	43.2g
↓ ÷12	↓ ÷1	↓ ÷16
4.06 mol	8.1 mol	2.7 mol
↓ ÷2.7	↓ ÷ 2.7	↓ ÷2.7
1.5	3	1

$C_3H_9O_2$: empirical formula.

(77g/mol)

$$n = \frac{226}{77} = 2.9 \approx 3.$$

$$(C_3H_9O_2)_{*3} = C_9H_{27}O_6$$

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ex) What is the empirical formula of a compound containing 18.7 % of (Li) 16.3 % of (C) and 65 % of (O).

ex) what is the empirical formula of a compound containing 67% of "C" 10.7% of "H" and 22.3 % of "O"

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Ch.3

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Balancing Chemical Equations

$$3A + 2B \longrightarrow C + 2D$$

Law of Conservation of energy

A, B : reactants.

C, D : products

3, 2, 1, 2 : coefficients

Ex. $N_{2\,(g)} + H_{2\,(g)} \longrightarrow NH_{3\,(g)}$

N_2 → 2 (N) atoms

H_2 → 2 (H) atoms

NH_3 → 1 (N) atom, 3 (H) atoms

$$1N_2 + 3H_2 \longrightarrow 2NH_3$$

Coefficients respectively are:

1, 3, 2

Coefficients could be fractions

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but usually coefficients in their simplest whole no. ratio.

ex. $_C_6H_6{}_{(\ell)} + _O_2{}_{(g)} \longrightarrow _CO_2{}_{(g)} + _H_2O{}_{(g)}$

$C_6H_6 + O_2 \longrightarrow 6CO_2 + H_2O$

- "Balance atoms else "H" and "O" first.

$C_6H_6 + O_2 \longrightarrow 6CO_2 + 3H_2O$

$C_6H_6 + \frac{15}{2} O_2 \longrightarrow 6CO_2 + 3H_2O$

fraction: so multiply the whole equation by "2"

$2C_6H_6 + 15O_2 \longrightarrow 12CO_2 + 6H_2O$

The simplest whole no. ratio is:

2, 15, 12, 6 respectively.

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Balance the following equations to simplest whole no. coefficients.

$BF_3 + Li_2SO_3 \longrightarrow B_2(SO_3)_3 + LiF$

coefficients: 2, 3, 1, 6 respectively.

ex) $P + Cl_2 \longrightarrow PCl_3$

$2P + 3Cl_2 \longrightarrow 2PCl_3$.

2 mol "P" $\Leftrightarrow$ 3 mol Cl_2 $\Leftrightarrow$ 2 mol PCl_3

$\Rightarrow$ mole ratio between reactants & products

$$\frac{\text{moles "P"}}{2} = \frac{\text{moles } Cl_2}{3} = \frac{\text{moles } PCl_3}{2}$$

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Ex. For the reaction: (combustion rxn)

$$C_3H_8 + O_2 \longrightarrow CO_2 + H_2O$$

Balance the equation and find mass of CO_2 (44 g/mol) produced for combustion of 3.782 g of C_3H_8 (44 g/mol).

$$C_3H_8 + 5O_2 \longrightarrow 3CO_2 + 4H_2O$$

$$\frac{\text{moles } C_3H_8}{1} = \frac{\text{moles } CO_2}{3}$$

$$\frac{(3.782/44)}{1} = \frac{(\text{mass}/44)}{3}$$

mass of CO_2 produced = 3.782×3

= 11.346 g CO_2.

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For the same rxn:

$$C_3H_8 + 5\,O_2 \longrightarrow 3CO_2 + 4\,H_2O$$

Find mass of water produced when 10g of CO_2 (44g/mol) is formed?

$$\frac{\text{moles } CO_2}{3} = \frac{\text{moles } H_2O}{4}$$

$$\frac{(10/44)}{3} = \frac{\text{moles } H_2O}{4}$$

$$\text{moles } H_2O = \frac{(10/44) \times 4}{3}$$

$$= 0.303 \text{ mol } H_2O$$

$$\text{mass } H_2O = 0.303 \text{ mol} \times 18 \text{ g/mol}$$

$$= 5.45 \text{ gram.}$$

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Limiting Reactant.

$3A + 2B \longrightarrow C$

At the end of the rxn ↓

1. 

   A = 0 mol
   B = 0 mol
   C = 1 mol

2. 

   A = 0 mol
   B = 0 mol
   C = 2 mol

3. 

   A = 1 mol remained
   B = 0 mol.
   C = 1 mol.

4. 

   A = 0 mol.
   B = 1 mol remained.
   C = 1 mol produced

Case (1), (2) : Both "A", "B" are limiting reactants.

Case "3" : "B" is L.R , "A" is excess reactant

Case "4" : "A" is L.R , "B" is excess.

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How to determine the L.R of a rxn:

The limiting reactant is the reactant that is consumed completely.

It determines amounts of all products.

ex. $3A + 2B \longrightarrow C.$

7mol 6mol ?

From equation : 3 mol A $\Leftrightarrow$ 2 mol B

7 mol $\Leftrightarrow$?

$\Rightarrow$ 4.67 mol of "B"

"B" is excess reactant

"A" is L.R. $\Rightarrow$ limiting reactant

Amount of "C" produced = 2.33 mol "C"

3 mol A $\Leftrightarrow$ 1 mol C

7 mol $\Leftrightarrow$?

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Thermodynamics. Lecture (14)

$$\Delta E = \Delta H - \underbrace{\Delta n_g R T}_{J}$$

Hess's Law:

The enthalpy of the overall process is the sum of enthalpy change of the steps in the process.

$$\Delta H_{rxn} = \Delta H_1 + \Delta H_2 + \Delta H_3 + \cdots$$

Remember.

If $A + 2B \longrightarrow 3C$ $\quad \Delta H = -10$ KJ

$2A + 4B \longrightarrow 6C$ $\quad \Delta H = -10 \times 2 = -20$ KJ

$\frac{3}{2} C \longrightarrow \frac{1}{2} A + B$ $\quad \Delta H = +5$ KJ.

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Given the following thermochemical equations:

(1) $H_{2(g)} + Cl_{2(g)} \longrightarrow 2HCl_{(g)}$ $\quad \Delta H_1 = -185\ KJ$

(2) $H_{2(g)} + Br_{2(g)} \longrightarrow 2HBr_{(g)}$ $\quad \Delta H_2 = -73\ KJ$

Find ΔH for the reaction

$$2HCl_{(g)} + Br_{2(g)} \longrightarrow 2HBr_{(g)} + Cl_{2(g)}$$

Equ (1) $(* -1)$ $\quad 2HCl \longrightarrow \cancel{H_2} + Cl_2$

Equ (2) $(* 1)$ $\quad \cancel{H_2} + Br_2 \longrightarrow 2HBr$

Net rxn : $2HCl_{(g)} + Br_{2(g)} \longrightarrow 2HBr_{(g)} + Cl_{2(g)}$

$\Delta H_{rxn} = (-185 \times -1) + (-73 \times 1) = +112\ KJ$

$= -\Delta H_1 + \Delta H_2 = +112\ KJ$

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Given the following thermochemical equ.

$C_{(s)} + 2H_{2(g)} \longrightarrow CH_{4(g)}$ $\Delta H_1 = -74.8$ KJ

$C_{(s)} + O_{2(g)} \longrightarrow CO_{2(g)}$ $\Delta H_2 = -394$ KJ

$2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(g)}$ $\Delta H_3 = -484$ KJ

$H_2O_{(\ell)} \longrightarrow H_2O_{(g)}$ $\Delta H_4 = 44$ KJ

Find ΔH for the reaction:

$$2CH_{4(g)} + 4O_{2(g)} \longrightarrow 2CO_{2(g)} + 4H_2O_{(\ell)}$$

Equ (1) (x -2) $2CH_4 \longrightarrow 4H_{2(g)} + 2C_{(s)}$

Equ (2) (x 2) $2C_{(s)} + 2O_{2(g)} \longrightarrow 2CO_{2(g)}$

Equ (3) ?

Equ (4) (x -4) $4H_2O_{(g)} \longrightarrow 4H_2O_{(\ell)}$

Equ (3) (x -2) $4H_{2(g)} + 4O_{2(g)} \longrightarrow 4H_2O_{(g)}$

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overall rxn:

$$2CH_{4\,(g)} + 4O_{2\,(g)} \longrightarrow 2CO_{2\,(g)} + 4H_2O_{(\ell)}$$

$$\Delta H_{rxn} = (\Delta H_1 \times -2) + (\Delta H_2 \times 2) + (\Delta H_3 \times -2) + (\Delta H_4 \times -4)$$

$$\Delta H_{rxn} = (-74.8 \times -2) + (-395 \times 2) + (-484 \times -2) + (44 \times -4)$$

$$\Delta H_{rxn} = +153.6 \text{ KJ} \quad \text{(endothermic rxn)}$$

$$\Delta H_{rxn} = 76.8 \text{ KJ / mol } CH_4$$

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Stoichiometery

Limiting reactant :

For the rxn :

$$C_6H_6 + \frac{15}{2} O_2 \longrightarrow 6\,CO_2 + 3H_2O$$

78 g/mol 32 g/mol 44 g/mol 18 g/mol

A. Calculate mass of CO_2 produced when 14.8 g of C_6H_6 are allowed to react with 22g of O_2.

- moles $C_6H_6 = \frac{14.8\,g}{78\,g/mol} = 0.189\,mol\ C_6H_6$

- moles $O_2 = \frac{22\,g}{32\,g/mol} = 0.688\,mol\ O_2$.

which is the limiting reactant ?

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From balanced equation:

$$1\ mol\ C_6H_6 \iff 7.5\ mol\ O_2$$

$$0.189\ mol \iff ?$$

$$\text{moles } O_2 \text{ needed} = \frac{0.189 \times 7.5}{1}$$

$$= 1.42\ mol\ O_2$$

$$\downarrow \times 32\ g/mol$$

$$\Rightarrow 45.36\ g\ O_2$$

should be existed to react with C_6H_6

Actually there is only 22g allowed to react. So "O_2" will all be consumed.

So "O_2" is the L.R.

"O_2" سوف تحدّد كميات النواتج جميعها.

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$\frac{\text{moles } C_6H_6}{1}$ Vs $\frac{\text{moles } O_2}{7.5}$

$\frac{0.189 \text{ mol}}{1}$ $\frac{0.688}{7.5}$

0.189 0.092

(Excess reactant) Smaller no. is the L.R

To find moles of CO_2 :

$$\frac{\text{moles of } O_2}{7.5} = \frac{\text{moles } CO_2}{6}$$

$$\frac{0.688}{7.5} = \frac{x}{6}$$

$x =$ 0.55 mol CO_2 $\xrightarrow{\times 44 \text{ g/mol}}$ mass CO_2

(24.2 g CO_2)

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b. If only 20g of CO_2 were actually produced, Calculate percentage yield for the rxn.

20g CO_2 actually produced. (actual yield) (الناتج الفعلي)

24.2g CO_2 should be produced according to Stoichiometric calculations (الناتج النظري) (Theoritical yield)

$$\text{Percentage yield (المردود المئوي)} = \frac{\text{actual yield (g or mol)}}{\text{Theoritical yield (g or mol)}} \times 100\%$$

$$\text{Percentage yield} = \frac{20\,g}{24.2\,g} \times 100\,\%$$

$$= 82.6\,\%$$

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For the rxn:

$S_2 + 3O_2 \longrightarrow 2SO_3$

64 g/mol 32 g/mol 80 g/mol.

S = 32
O = 16

when 10g S_2 were allowed to react with 5g of O_2, 5.71 g of SO_3 were produced.

a. Calculate theoritical yield of SO_3

$$\frac{\text{moles } O_2}{3} = \frac{\text{moles } SO_3}{2}$$

$$\frac{(5/32)}{3} = \frac{(\text{mass}/80)}{2}$$

mass of SO_3 = 8.33 g

theoritical yield.

which is L.R ?

S_2	O_2
$\frac{(10/64)}{1}$	$\frac{(5/32)}{3}$
0.156	0.05 الأصغر L.R

b. Calculate percentage yield.

$$= \frac{\text{actual } SO_3}{\text{theoritical } SO_3} \times 100\% = \frac{5.71}{8.33} \times 100\%$$

$$= 68.5\%$$

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Lecture (15)

Standard heat of formation (ΔH_f°)

حرارة تكوين المواد المعيارية

"Heat associated with formation of one mole of substance from its elements in their standard states.

Standard states of elements (25°C, 1atm)

Gases	liquids	Solids
(1) Noble elements $He_{(g)}$ $Ne_{(g)}$ $Ar_{(g)}$ $Kr_{(g)}$	$Hg_{(\ell)}$ $Br_{2(\ell)}$	IA : $Na_{(s)}$, ... IIA : $Mg_{(s)}$, ... IIIA : $Al_{(s)}$, ... Transtion elements $Cu_{(s)}$, $Pt_{(s)}$, $Fe_{(s)}$ $Cr_{(s)}$, ... $P_{4(s)} \Rightarrow P_{(s)}$ $S_{8(s)} \Rightarrow S_{(s)}$ $C_{(s)} \Rightarrow$ graphite. $I_{2(s)}$
(2) Some nonmetals $H_{2(g)}$ $N_{2(g)}$ $O_{2(g)}$ $F_{2(g)}$ $Cl_{2(g)}$		

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Values of ΔH_f° of substances are all tabulated in (KJ/mol)

Substance	ΔH_f° (KJ/mol)
$CH_{4\,(g)}$	-74.8
$H_2O_{\,(\ell)}$	-286
$H_2O_{\,(g)}$	-242
$CO_{2\,(g)}$	-394
$O_{2\,(g)}$	
$C_2H_{2\,(g)}$	226.7
$C_2H_5OH_{\,(\ell)}$	-278

Writing formation equations:

$C_{(s)} + O_{2\,(g)} \longrightarrow CO_{2\,(g)}$ $\quad \Delta H^\circ = -394$ KJ

$H_{2\,(g)} + \frac{1}{2}O_{2\,(g)} \longrightarrow H_2O_{(\ell)}$ $\quad \Delta H = -286$ KJ

$\frac{1}{2}H_{2\,(g)} + \frac{1}{2}Br_{2\,(\ell)} \longrightarrow HBr$ $\quad \Delta H = \Delta H_{f\,(HBr)}^\circ$

$= -36.4$ KJ

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Which of the following reactions is a standard formation reaction

$N_{2\,(\ell)} + O_{2\,(g)} \longrightarrow N_2O_{4\,(\ell)}$ ✗

$Cl_{2\,(g)} + H_{2\,(g)} \longrightarrow 2HCl_{(\ell)}$ ✗

$CO_{(g)} + 2H_{2\,(g)} \longrightarrow CH_3OH_{(\ell)}$ ✗

$2Fe_{(s)} + 3O_{(g)} \longrightarrow Fe_2O_{3\,(s)}$ ✗

$2Hg_{(\ell)} + Cl_{2\,(g)} \longrightarrow Hg_2Cl_{2\,(s)}$ ✓

$\frac{1}{2}H_{2\,(\ell)} + \frac{1}{2}I_{2\,(g)} \longrightarrow HI$ ✗

$C_{(s)} + \frac{1}{2}O_{2\,(g)} + 2H_{2\,(g)} \longrightarrow CH_3OH_{(\ell)}$ ✓

$HNO_{3\,(\ell)} \longrightarrow \frac{1}{2}H_{2\,(g)} + \frac{3}{2}O_{2\,(g)} + \frac{1}{2}N_{2\,(g)}$ ✗

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Calculating ΔH_{rxn} using values of ΔH_f°

$$\Delta H_{rxn} = \sum n\,\Delta H_f^\circ{}_{(products)} - \sum n\,\Delta H_f^\circ{}_{(reactants)}$$

n: coefficients for each reactant and product.

For the reaction: $3A + 2B \rightarrow C + \frac{1}{2}D$

Given ΔH_f° values: in (KJ/mol)

ΔH_f° (A) = -50

ΔH_f° (B) = 30

ΔH_f° (C) = -70

ΔH_f° (D) = -100

$$\Delta H^\circ_{rxn} = \left[\left(\frac{1}{2}\text{ mol} \times -100\ \text{KJ/mol}\right) + (1 \times -70)\right] - \left[(3 \times -50) + (2 \times 30)\right]$$

$$\Delta H_{rxn}^\circ = -30\ \text{KJ}.$$

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Calculate standard heat of formation of C_2H_5OH given that:

$\Delta H_f^\circ \quad H_2O_{(\ell)} = -286 \quad KJ/mol$

$\Delta H_f^\circ \quad CO_{2\,(g)} = -394 \quad KJ/mol$

$$C_2H_5OH + 3O_{2\,(g)} \longrightarrow 2CO_{2\,(g)} + 3H_2O_{(\ell)}$$

If standard heat of combustion of ethanol = -1409 KJ.

$$\Delta H_{rxn}^\circ = -1409 \text{ KJ}.$$

$$\left[(3 \times -286) + (2 \times -394)\right] - \left[(3 \times 0) + 1 \times \Delta H_f^\circ\right] = -1409 \text{ KJ}$$

$$\Delta H_f^\circ \ (C_2H_5OH) = -237 \text{ KJ/mol}.$$

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Examples:

Which of the following has $\Delta H_f^\circ \neq 0$?

$C_{(s)}$

$O_{2(g)}$

$H_{2(\ell)}$ ← $H_{2(g)}$

$P_{(s)}$

$F_{2(g)}$

$Ar_{(g)}$

$2Br_{(\ell)}$ ← $Br_{2(\ell)}$

$I_{2(s)}$

$Cu_{(\ell)}$ ← $Cu_{(s)}$

- Write the standard formation equation of $H_2SO_{4(\ell)}$

$$H_{2(g)} + 2O_{2(g)} + S_{(s)} \longrightarrow 1\ H_2SO_4$$

- Given the thermochemical equations.

$$2H_{2(g)} + O_{2(g)} \longrightarrow 2H_2O_{(\ell)} \quad \Delta H = -136.6\ KJ/mol$$

$$N_2O_{5(g)} + H_2O_{(\ell)} \longrightarrow 2HNO_{3(\ell)} \quad \Delta H = -18.3\ KJ/mol$$

$$\tfrac{1}{2}N_{2(g)} + \tfrac{3}{2}O_{2(g)} + \tfrac{1}{2}H_{2(g)} \longrightarrow HNO_{3(\ell)} \quad \Delta H = -41.6\ KJ/mol$$

Find ΔH_f° of N_2O_5 (in KJ/mol)

$= +6.8$ KJ/mol

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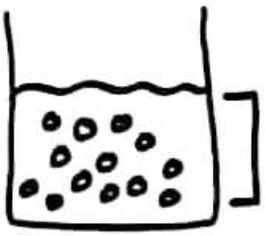
Ch.4 Chemical Solutions.

- Solute: مذاب
- Solvent: مذيب
- Solution: محلول

- Solution: homogeneous mixture of solute and solvent.

- Aqueous solution ⇒ Solvent is water.

ex. $HCl_{(aq)}$, $NaOH_{(aq)}$ ← محلول مائي



volume. = 1L.

5 mol : 1L.

100g : 1L

100g : 1000g

⇓ ⇓

Solute solution

هذه التغير عند تركيز المحلول

10g : 1Kg

↓

Solute : Solvent

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Way to express concentration of a solution:

- Remember: Concentration is a ratio between $\left(\frac{\text{solute}}{\text{solution}}\right)$ or $\left(\frac{\text{solute}}{\text{solvent.}}\right)$

① Molarity $= \frac{\text{moles solute (mol)}}{\text{Liters solution}}$

$$\text{mol/L} = \text{mol.L}^{-1} \equiv \text{M (molar)}$$

② $(\text{w/w})\% = \frac{\text{mass solute (g)}}{\text{mass solution (g)}} \times 100\%$

ex) 5g NaCl in 0.5L water.

- density $(H_2O) = 1\ \text{g/mL}$
- mass (H_2O) (g) $=$ volume (H_2O) (mL)

$$\%\ \text{NaCl} = \frac{5\text{g}}{505\text{g}} \times 100\% = 1\%$$

(وزنه المذاب + المذيب) كـ
- وزنه المحلول .

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Ch.4 Chemical Solutions

- A solution is a mixture of solute and solvent

- A solution has a volume (L, mL), density (g/mL), mass (g)

  but does not have moles. or molar mass

- Concentration of Solutions:

1) Molarity = $\frac{\text{moles solute}}{\text{Liters of solution}}$ mol/L ≡ M

ex) Calculate molar concentration of a solution prepared by dissolving 2.715 g of KCl (= 74.5 g/mol) in 600 mL water:

$$M = \frac{\text{moles KCl}}{\text{V of soln (L)}} = \frac{\left(\frac{2.715}{74.5}\right)}{0.60\,L} = 0.06\ M$$

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- In molarity

Volume (flask) $\cong$ V (water) = V solution.

V: Volume.

- Calculate grams of KCl required to be dissolved in water in a 250mL flask with 0.32M concentration.

$$\text{moles (KCl)} = M_{soln} \cdot V_{soln.}$$

$$= 0.32M * 0.25L$$

$$= 0.08 \text{ mol KCl}$$

$$= 5.96 \text{ gram.}$$

How many grams of KCl are found in a 6.5 % KCl solution given that Volume of solution = 300mL and density of solution = 1.13 g/mL

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6.5 % ⇒ 6.5 g KCl ⇔ 100 g Solution.

Solution ↗ 300mL
↘ density = 1.13 g/mL

mass (soln) = d x V = $1.13 \frac{g}{mL}$ x 300mL

= 339 gram. soln.

6.5g KCl ⇔ 100 g soln.

? ⇔ 339 g soln

= 22.035 g KCl.

② % solute = w/w % = $\frac{\text{mass solute.}}{\text{mass soln}}$ x100 %

③ Molality = $\frac{\text{moles solute}}{\text{Kilograms solvent}}$ $\frac{mol}{kg} = m$ ↓ (molal)

$d_{(solvent)} = \frac{\text{mass (solvent)}}{V\ (solvent)}$

لتحويل الكتله
إلى حجم
أو العكس .

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Entropy "S"

Lecture "16"

A state function that describes the degree of disorder of a system.

$$\Delta S = S_f - S_i$$

Factors affecting entropy :

1) Temperature : As temp increase, randomness increase.

2) Physical state:

$$S_{(solid)} < S_{(liquid)} \ll S_{(gas)}$$

3) Volume of container (for gases)

As volume increase, entropy of gas molecules increase

4) Type of reaction.

Formation : $Ca + O_2 \longrightarrow CaO$ (decrease)

Dissociation : $NaCl \longrightarrow Na^+ + Cl^-$ (increase)

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Second Law of thermodynamics

"Entropy of universe increases for any spontaneous process."

$\Delta S_{univ} > 0$ for a spontaneous process.

Spontaneous process: (عمليه حدثت تلقائياً)

$$\Delta S_{rxn} = \frac{q}{T} \quad (J/K)$$

$$\Delta S^{\circ}_{rxn} = \frac{q}{298\ K.}$$

- For a spontaneous process:

$$\Delta H - T\Delta S < 0 \quad (-ve)$$

- For a nonspontaneous process:

$$\Delta H - T\Delta S > 0 \quad (+ve)$$

- For a reaction at equilibrium:

$$\Delta H - T\Delta S = 0$$

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Gibb's free Energy (G)

A state function that describes whether the reaction is spontaneous or not.

ΔG_{rxn} → (-ve) ⇒ Spontaneous rxn

→ (+ve) ⇒ nonspontaneous rxn

→ (= 0) ⇒ at equilibrium.

$A + B \rightleftharpoons C$.

$$\Delta G_{rxn} = \Delta H - T\Delta S$$

$$\Delta G^\circ_{rxn} = \Delta H^\circ - T\Delta S^\circ$$

ΔH° → KJ

ΔS° → J/K.

$T\Delta S^\circ$ ⇒ at standard conditions. (T = 298 K)

- ΔG_{rxn} = maximum work of the rxn.
- ΔG ⇒ depends on temperature.

ex) $H_2O_{(s)} \rightleftharpoons H_2O_{(\ell)}$ spont at 25°C.

(but nonspont at -12°C.)

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Predict if rxn is spont or not from signs of (ΔH) and (ΔS)

$$\Delta G = (\Delta H) + (-T\Delta S)$$

remember "T" is always (+ve)

ΔH	ΔS	ΔG	Notes
(−)	(+)	(−)	Reaction is spontaneous at all temperatures
(+)	(−)	(+)	Reaction is nonspont. at all temperatures.
(−)	(−)	(−)	Reaction is spont. at low temperatures $T < T_{eq}$
		(+)	Rxn is nonspont at: $T > T_{eq}$.
(+)	(+)	(−)	Rxn spont. at high temperatures $T > T_{eq}$.
		(+)	Nonspont at $T < T_{eq}$

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Predicting sign of ΔS_{rxn} :

- If Δn_g (of rxn) is (+ve) $\Rightarrow$ "S" increases

  ΔS_{rxn} : +ve

- If Δn_g (of rxn) is (-ve) $\Rightarrow$ "S" decreases

  ΔS_{rxn} : -ve.

- If $\Delta n_g = 0 \Rightarrow \Delta S \neq 0$. " not always zero"

Predict sign of ΔS_{rxn} for the following:

- $NaCl_{(s)} \longrightarrow Na^+_{(aq)} + Cl^-_{(aq)}$ — ΔS: +ve
- $H_2O_{(s)} \longrightarrow H_2O_{(g)}$ — ΔS: -ve
- $2NH_{3(g)} \longrightarrow N_{2(g)} + 3H_{2(g)}$ — $\Delta n_g = 4 - 2 = +2$, ΔS: (+ve)
- $Hg_{(\ell)} + I_{2(s)} \longrightarrow Hg_2I_{2(s)}$ — ΔS: -ve.
- $C_6H_{6(\ell)} + \frac{15}{2}O_{2(g)} \longrightarrow 6CO_{2(g)} + 3H_2O_{(g)}$

  $\Delta n_g = 9 - 7.5 = +1.5 \Rightarrow$ ΔS : +ve.

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Examples:

(1) For the reaction:

$$H_{2\,(g)} + \frac{1}{2} O_{2\,(g)} \longrightarrow H_2O_{(\ell)}$$

$$\Delta H_{rxn}^{\circ} = -286 \text{ KJ}$$

Is this rxn spontaneous or not?

ΔS sign is (-ve)

$\Delta n_g = 0 - 1.5 = (-1.5)$

ΔH sign is (-ve)

This reaction is spontaneous at low temperatures.

Note: at equilibrium $\Delta G = 0$.

$$\Delta G = \Delta H - T\Delta S = 0.$$

$$\Delta H - T_{eq}\,\Delta S = 0.$$

$$T_{(equ)} = \frac{\Delta H}{\Delta S}$$

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ex) For the reaction:

$$2A + B \longrightarrow C$$

If $\Delta H^\circ = +60$ KJ.

$\Delta S^\circ = 191 \frac{J}{K}$

At what temperatures will this rxn be spontaneous?

$$T_{eq} = \frac{\Delta H}{\Delta S} = \frac{60\ KJ}{0.191\ \frac{KJ}{K}} = 314.1\ K$$

(41.1°C)

The rxn is spont. at temperatures above 41.1°C.

(nonspont. at temps below 41.1°C)

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Phase Change (equilibrium systems).

Phase change	Name:	Temperature:	Calculation:
Solid → liquid	fusion	melting point (T_m)	$T_m = \frac{\Delta H_{fus}}{\Delta S_{fus}}$
liquid → gas	Vaporization	boiling point (T_b)	$T_b = \frac{\Delta H_{vap}}{\Delta S_{vap.}}$
liquid → Solid	freezing	Freezing point (T_f)	$T_f = \frac{\Delta H_{frez}}{\Delta S_{freez}}$
Solid → gas (directly)	Sublimation	Sublimation point (T_s)	$T_{sub} = \frac{\Delta H_{sub}}{\Delta S_{sub}}$

Example: Calculate melting point of I_2 if enthalpy of fusion $\Delta H_{fus} = 7.824\ KJ$ and entropy of fusion $\Delta S_{fus} = 124.5\ KJ/K$

$$T_m = \frac{\Delta H}{\Delta S} = \frac{7.824\ KJ}{0.124.5\ KJ/K} = 63\ K$$

$$= -209.9\ ^\circ C.$$

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Third Law of thermodynamics

Entropy for a pure solid crystalline material equals zero at zero Kelvin

$$S = 0 \quad \text{at} \quad 0\ K$$

Chemical compounds

- $\Delta H_f^\circ \Rightarrow KJ/mol$
- $\Delta G_f^\circ \Rightarrow KJ/mol$
- $S^\circ \Rightarrow J/mol.K$

- Values of ΔH_f°, ΔG_f°, S° are given in tables for compounds.

- For elements in their standard states:

$$\Delta H_f^\circ = 0$$

$$\Delta G_f^\circ = 0$$

$$S^\circ \neq 0.$$

للعناصر في حالتها المعيارية ⇐

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Given the following thermochemical values:

Substance	ΔH_f° (KJ/mol)	S° (J/mol.K)	ΔG_f° (KJ/mol)
$C_6H_{6\,(\ell)}$	49	172.8	124.5
$O_{2\,(g)}$	0	205.14	0
$CO_{2\,(g)}$	-394	213.7	-394
$H_2O_{(g)}$	-242	188.8	-229
$H_2O_{(\ell)}$	-286	70	-237

For the rxn:

$$C_6H_{6\,(\ell)} + \frac{15}{2}O_{2\,(\ell)} \longrightarrow 6CO_{2\,(g)} + 3H_2O_{(\ell)}$$

$$\Delta H^\circ_{rxn} = \sum n\,\Delta H_f^\circ{}_{(products)} - \sum m\,\Delta H_f^\circ{}_{(reactants)}$$

$$\Delta S^\circ_{rxn} = \sum n\,S^\circ_{(products)} - \sum m\,S^\circ_{(reactants)}$$

$$\Delta G^\circ_{rxn} = \sum n\,\Delta G_f^\circ{}_{(products)} - \sum m\,\Delta G_f^\circ{}_{(reactants)}$$

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$$\Delta H^\circ_{rxn} = [(6 \times -394) + (3 \times -286)] - [(7.5 \times 0) + (1 \times 49)]$$

$$= -3271 \text{ KJ}$$

$$\Delta S^\circ_{rxn} = [(6 \times 213.7) + (3 \times 70)] - [(7.5 \times 205) + (1 \times 172.8)]$$

$$= -218.1 \text{ J/K}$$

$$\Delta G^\circ_{rxn} = [(6 \times -394) + (3 \times -237)] - [(7.5 \times 0) + (1 \times 124.5)]$$

$$= -3199.5 \text{ KJ} \approx -3200 \text{ KJ}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$= (-3271 \text{ KJ}) - (298 \text{ K} \times -218.1 \text{ J/K})$$

$$= -3206 \text{ KJ}$$

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Chemical Kinetics:

rate of reaction $= \frac{\Delta \text{Conc.}}{\Delta \text{time}}$ (M/s) ← unit of rate.

($mol.L^{-1}.s^{-1}$)

$$2A + 3B \rightarrow \frac{1}{2}C + D$$

rate of disappearance of "A" $= -\frac{\Delta[A]}{\Delta t}$

rate of disappearance of "B" $= -\frac{\Delta[B]}{\Delta t}$

rate of appearance of "C" $= \frac{\Delta[C]}{\Delta t}$

rate of appearance of "D" $= \frac{\Delta[D]}{\Delta t}$

- rate of reaction $=$

$$\frac{-\Delta[A]}{2\Delta t} = \frac{-\Delta[B]}{3\Delta t} = \frac{\Delta C}{\frac{1}{2}\Delta t} = \frac{\Delta D}{\Delta t}$$

$$\frac{\text{rate of A}}{2} = \frac{\text{rate of B}}{3} = 2.\text{rate of C} = \text{rate of D}$$

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Remember: $2A + 3B \rightarrow \frac{1}{2}C + D$

$$A = -\frac{\Delta[A]}{\Delta t} \qquad B = -\frac{\Delta[B]}{\Delta t}$$

$$C = \frac{\Delta[C]}{\Delta t} \qquad D = \frac{\Delta[D]}{\Delta t}$$

So: $\frac{A}{2} = \frac{B}{3} = 2C = D$

Which is correct:

- rate of $A = \frac{3}{2}$ rate of B incorrect
- rate of "C" $= \frac{2}{3}$ rate of "B" incorrect
- $\frac{\Delta[A]}{\Delta t} = 4\frac{\Delta[C]}{\Delta t}$ incorrect
- $\frac{\Delta[B]}{\Delta t} = \frac{3}{2}\frac{\Delta[A]}{\Delta t}$ Correct
- $\frac{\Delta[C]}{\Delta t} = -\frac{1}{4}\frac{\Delta[A]}{\Delta t}$ Correct

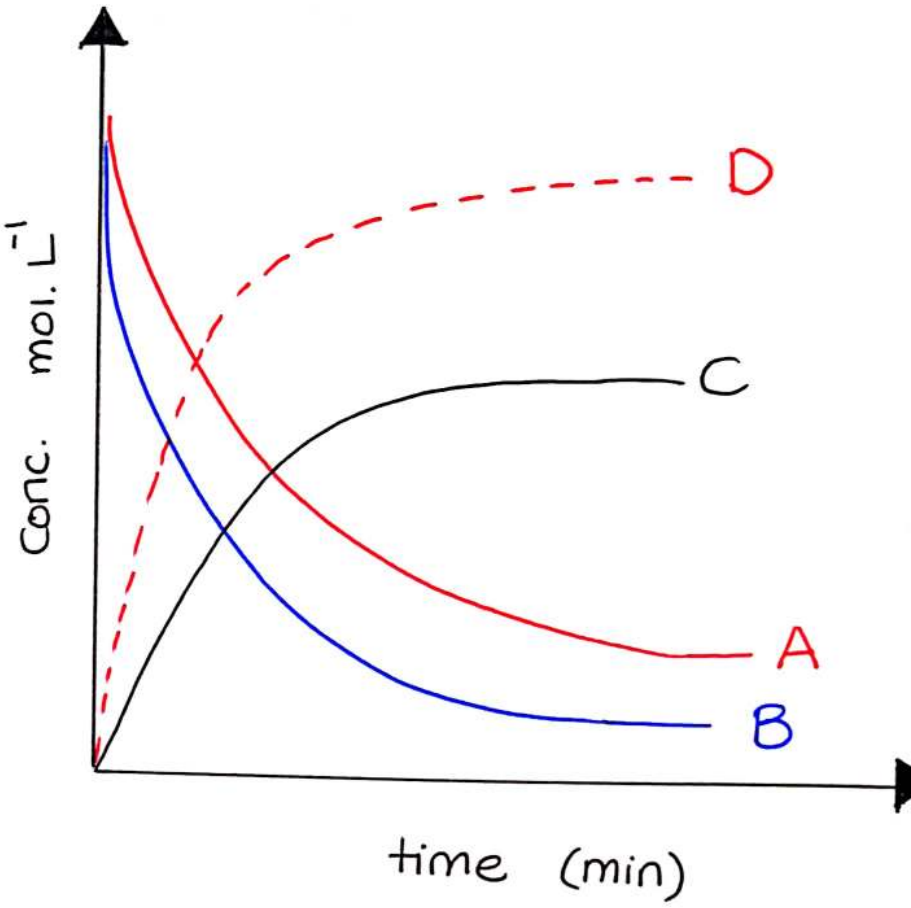
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Average rate of reaction =

$$-\frac{\Delta[\text{reactant}]}{\text{Coeff.} \times \Delta t} = \frac{\Delta[\text{product}]}{\text{Coeff.} \times \Delta t} \quad (M/s)$$

Δ Conc. $\equiv$ Final conc. - initial conc.

- Rate of rxn is always (+ve)



- Concentrations of "A", "B" decreases with time.
- Concentrations of "C", "D" increases with time.

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Maximum rate of reaction will be at the beginning of the reaction.

$$2A + B \longrightarrow C$$

Time (min)	Concentration (M)	Average rate (M/s)
0	2.00	$-\frac{[1.67-2]}{2\times1\times60} = 1.98$ M/s
1	1.67	$-\frac{[1.39-1.67]}{2\times1\times60} = 2.3\times10^{-3}$
2	1.39	$-\frac{[1.17-1.39]}{2\times1\times60} = 1.83\times10^{-3}$
3	1.17	
4	0.97	

$$\text{rate of reaction} = \frac{-\Delta[A]}{2\,\Delta t}$$

($\Delta[A]$ → (M), Δt → (s))

- Calculate rate of consumption of "A" after 4 minutes.

$$\text{rate of "A"} = \frac{-(0.97-2)}{4\times60} = 4.3\times10^{-3}\ \text{M/s}$$

$$\text{rate of rxn} = \frac{-(0.97-2)}{2\times4\times60} = 2.145\times10^{-3}\ \text{M/s}$$

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Suppose: you have a 13.7% of HCl solution. Calculate its molar concentration given that:

- density of HCl solution = 1.41 g/mL.
- F.wt of HCl = 36.5 g/mol

13.7 % HCl → 13.7 g HCl (solute)
→ 100g solution.

$$\text{Molarity} = \frac{\text{moles HCl}}{\text{V of soln (in L)}}$$

$$\text{Molarity} = \frac{\left(\text{mass}/\text{F.wt}\right) \Rightarrow \text{Solute : HCl}}{\left(\text{Mass}/\text{density}\right) \Rightarrow \text{Solution.}}$$

$$\text{Molarity} = \frac{(13.7\text{g}/36.5)}{(100\text{g}/1.41\ \text{g/mL})} = \frac{0.375\ \text{mol}}{70.92}$$

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$$\text{Molarity} = \frac{0.375\ \text{mol}}{(70.92\ \text{mL} / 1000)\ \text{L}}$$

$$= 5.28\ \text{M}.$$

$$\text{Molarity} = \frac{\% \times \text{density} \times 10}{\text{F.wt}}$$

$$= \frac{13.7 \times 1.41 \times 10}{36.5} = 5.29\ \text{M}.$$

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General Chemistry Lecture "6" (online)

- Molarity = $\frac{\% \times \text{density} \times 10}{\text{molar mass}}$.

- Suppose you have:

[50mL. 0.2M HCl soln (تركيز أعلى)] ⇒ [0.1M HCl (تركيز أقل)]

⇒ you can add a certian amount of water.

⇒ moles HCl (before adding water) = moles HCl (after adding water)

- $M_1V_1 = M_2V_2$ ⇒ "للمحلول نفسه"

M_1V_1: concentrated soln المركز

M_2V_2: diluted soln المحلول المخفف

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M_1 : molarity of soln before dilution.

M_2 : molarity of soln after dilution.

V_1 : volume before adding water.

V_2 : volume after adding water.

$V_2 > V_1$ $\quad V_2 - V_1 = V_{water}$. $\quad M_1 > M_2$.

ex).

$\Rightarrow$ 50mL 0.2M HCl $\xrightarrow[\text{water.}]{\text{add}}$ 0.1M HCl, V = ?

$$M_1V_1 = M_2V_2.$$

$$0.2 \times 50\text{mL} = 0.1\text{M} \times V_2$$

$$V_2 = 100\text{mL}.$$

So water added $= V_2 - V_1$

$= 100\text{mL} - 50\text{mL}$

$= 50\text{mL}$ water.

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ex) How many milliliters of 0.35 M to prepare a 250mL of 0.24M of NaOH solution.

$$M_1 V_1 = M_2 V_2.$$

$$0.35M \times V_1 = 0.24M \times 250mL$$

$$V_1 = 171.4\ mL.$$

$$V\ (\text{water added}) = 250 - 171.4\ mL$$

$$= 78.6\ mL\ \text{water}.$$

ex) Calculate the volume of water that should be added to a 100mL (0.42M) to reduce its concentration to one third its original conc.

V_1: 100mL

M_1: 0.42M

$\xrightarrow[\text{water.}]{?}$ $(\frac{1}{3} \times 0.42M) = M_2$

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$$M_1 V_1 = M_2 V_2.$$

$$V_2 = \frac{0.42\,M \times 100\,mL}{\frac{1}{3} \times 0.42} = 300\,mL.$$

$$V_{water} = 300 - 100 = 200\,mL \text{ water.}$$

- Concentration of Ions:

$0.2M\ CaCl_2$ →
- $0.2M \times 1 = [Ca^{2+}]$
- $0.2M \times 2 = [Cl^-]$

$$\left\{ \begin{matrix} 1L \\ 0.2mol \end{matrix} \right\}$$

$$[Ion] = M_{compound} \times \text{subscript.}$$

$$M_{Ca_3(PO_4)_2} = 0.3M.$$

$$[Ca^{2+}] = 0.3M \times 3 = 0.9M$$

$$[PO_4^{3-}] = 0.3M \times 2 = 0.6M$$

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Ch.4 "Chemical Solutions"

Lecture (7)
"online."

Reactions of solutions:

Types of Chemical reactions:

1) Acid - Base reactions (neutralization rxn)

$$NaOH_{(aq)} + HCl_{(aq)} \longrightarrow NaCl_{(aq)} + H_2O_{(\ell)}$$

2) Precipitation reactions:

$$AgNO_{3\,(aq)} + NaCl_{(aq)} \longrightarrow NaNO_{3\,(aq)} + AgCl_{(s)}$$

3) Reduction - oxidation rxns: (Redox rxns)

$$Fe^{2+}_{(aq)} + MnO_4^-{}_{(aq)} \xrightarrow{OH^-} Fe^{3+}_{(aq)} + MnO_{2\,(aq)}$$

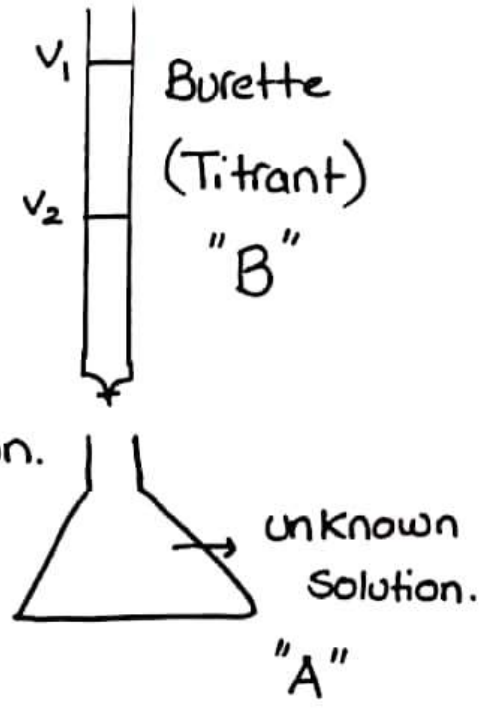
- The balanced equation is the base of stoichiometric calculations.

- moles = $M \times V_{(L)}$ ⇐ لحساب مولات مذاب في محلول

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Titration : المعايرة

A way or technique used to find out moles (or molarity) of a solution by reacting with an solution of exact (Known) molarity called titrant.



- An indicator is usually added to solution before starting titration to get the end point of titration.

- A balanced chemical equation is needed.

$$2A + B \longrightarrow C + D.$$

- At end point "where color changes"

$$\frac{\text{moles of "A"}}{2} = \frac{\text{moles of "B"}}{1.}$$

$$\frac{(M \times V)_A}{2} = \frac{(M \times V)_B}{1}$$

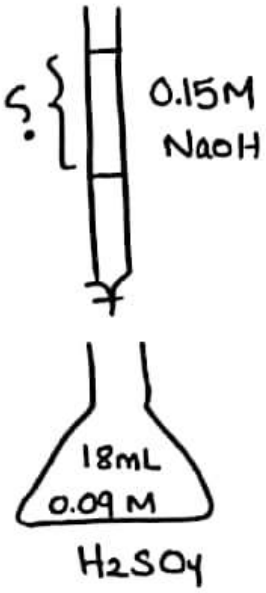
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Example: For the reaction.

$$2NaOH_{(aq)} + H_2SO_{4\,(aq)} \longrightarrow Na_2SO_{4\,(aq)} + 2H_2O_{(\ell)}$$

Calculate milliliters of 0.15M NaOH required to neutralize (completely react) with 18mL (0.09M) of H_2SO_4.

$$\frac{(M \times V)_{NaOH}}{2} = \frac{(M \times V)_{H_2SO_4}}{1}$$



$$\frac{0.15M \times V_{NaOH}}{2} = \frac{0.09 \times 18mL}{1}$$

$$V_{NaOH} = \frac{0.09M \times 18mL \times 2}{0.15\,M}$$

$$V_{NaOH} = \boxed{21.6\ mL}\ NaOH.$$

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Ex. For the reaction:

$$3Ag_2(SO_4)_{(aq)} + 2AlBr_{3\,(aq)} \longrightarrow 6AgBr_{(s)} + Al_2(SO_4)_{3\,(aq)}$$

When mixing 56mL (0.06M) of Ag_2SO_4 with 75 mL (0.11 M) of $AlBr_3$ in a rxn vessel. Answer the following:

(a) Calculate mass of AgBr. (187.7 g/mol) precipitated at the end of rxn.

moles of Ag_2SO_4 = $M \times V_{(L)}$

$= 0.06 \times \frac{56}{1000}$ L $= 3.36 \times 10^{-3}$ mol

moles of $AlBr_3$ = $M \times V_{(L)}$

$0.11M \times \frac{75}{1000}$ L = 8.25×10^{-3} mol.

Which is the L.R ?

$Ag_2SO_4 \Rightarrow 3.36 \times 10^{-3} / 3 = 1.1 \times 10^{-3}$ ← الأصغر

$AlBr_3 \Rightarrow 8.25 \times 10^{-3} / 2 = 4.1 \times 10^{-3}$

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Ag_2SO_4 is the L.R.

$$\frac{\text{moles of } AgBr}{6} = \frac{\text{moles of } Ag_2SO_4}{3}$$

$$\frac{x}{6} = \frac{3.36 \times 10^{-3}}{3}$$

x: moles of AgBr precipitate $= 6.72 \times 10^{-3}$ mol

$\downarrow \times 187.7$ g/mol

1.26 gram
(ترسبت في المحلول)

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Ch.6 : Thermodynamics.

This chapter deals with energy associated to chemical reactions.

Important terms:

- system : النظام
- Surroundings : المحيط ⇒ universe = sys + surr.
- Universe : الكون

- Law of Conservation of energy:

"Energy is neither created nor destroyed but changes from one form to another."

- $E_{universe}$ is constant.
- $\Delta E_{universe} = 0$.
- $\Delta E_{sys} + \Delta E_{surr} = 0$
- $\Delta E_{sys} = -\Delta E_{surr}$.

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- q: heat of reaction (or system).
- w: work done by or on system
- Both have units of J, KJ.

1) Heat (q)

q_{sys}
- $+$ ⇒ system absorbs heat
- $-$ ⇒ system evolves heat.

- absorb = gain = add يكسب
- evolve = release = lose يخسر

2) Work (w)

$w_{(sys)}$
- $+$ ⇒ system perform work on system.
- $-$ ⇒ system performs work on surrounding.

+ve work: work done on system.

-ve work: work done by system

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First Law of Thermodynamics:

$$\Delta E_{sys} = q + w.$$

ex) For a chemical system which absorbs 70 J of heat while performing 100 J of work.

Calculate the change in its internal energy.

$$\Delta E_{sys} = q + w.$$

$$= (+70)J + (-100)J$$

$$= -30\,J.$$

$$\Delta E_{surr} = +30\ J.$$

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ex) $A + 2B \longrightarrow C + 50\,KJ$

- Exothermic rxn $\Rightarrow$ $(q = -50\ KJ.)$

$A + 2B + 50\,KJ \longrightarrow C.$

- Endothermic rxn. $\Rightarrow$ $(q = +50\ KJ)$

* E (internal energy) is a state function.

* State function means: it depends on the initial state and the final states only, regardless the pathway of reaction.

$A + 2B \longrightarrow C$

(initial state) reactants

(final state) products.

$\Delta E_{rxn} = E_f - E_i$

$= E_{products} - E_{reactants}.$

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State function :

One pathway is :

1) $A + 2B \longrightarrow C.$

initial state — final state.

$\Delta E = q + w.$

$= -80\ KJ.$

$q = -60.\ KJ$

$w = -20\ KJ.$

Another pathway is :

2) $A + D \longrightarrow E$ step (1)

$E + 2B \longrightarrow D + C.$ step (2)

$A + 2B \longrightarrow C \Rightarrow \Delta E = -80\ KJ.$

initial state — final state

$q = +30.\ KJ$

$w = -110\ KJ.$

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Types of systems:

① open system : ("P" is constant)

Mass and heat transfer is allowed between system and surroundings.

② closed system : (Volume is constant)

heat transfer is allowed

mass transfer is not allowed.

③ Isolated system:

No heat transfer.

No mass transfer.

④ Isothermal system: (T is constant)

temperature is held constant.

⑤ Adiabatic system:

No heat transfer.

Mass transfer is allowed.

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Ch.6 Thermodynamics. Lecture "9" online

First Law of Thermodynamics.

$$\Delta E_{sys} = q + w$$

- The work to be calculated is a type of work called PV- work.
- PV- work is work related to gaseous systems

$$w = -P_{ext} \cdot \Delta V$$

↓

For a gas expanded or compressed in a cylinder.

$$w = -\Delta n_g RT$$

↓

For a chemical rxn that involving gases (as reactants and/or products)

- P: pressure.
- V: volume.
- n: moles of gas.
- R: universal gas constant
- T: temperature.

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PV- work.

$w = -P_{ext} \cdot \Delta V$

For expansion or compression of a gas inside a cylinder.

P_{ext} : pressure in KPa

$\Delta V = V_f - V_i$ (in L)

KPa . L = 1 J.

$w = -\Delta n_g . R . T$

For a chemical rxn that has gaseous reactants or/and products.

R: universal gas constant

$= 0.0821 \text{ atm.L mol}^{-1}.\text{K}^{-1}$

$= 8.314 \text{ J/mol.K}$

T: temperature in (Kelvin)

Δn_g = (moles of gaseous products) − (moles of gaseous reactants)

$2A_{(g)} + B_{(aq)} \longrightarrow C_{(aq)} + D_{(g)}$

$\Delta n_g = (1) - (2)$

$= -1$ mol.

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P: 1atm = 101.3KPa = 760mmHg

T: ___K = ___°C + 273

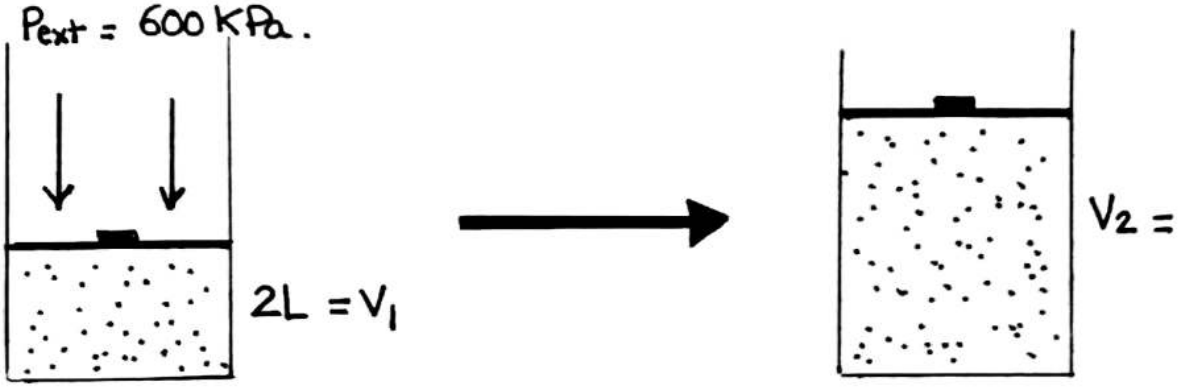
V: 1L = 1 dm^3 = 1000 mL = 1000 cm^3

PV- work

(1) A gas in a cylinder → Expansion
→ Compression

"غاز محصور داخل اسطوانه وعليه مكبس قابل للحركه تمدد الغاز أو تقلصه يعتمد على فرق الضغط بين الضغط الخارجي (على المكبس) وضغط الغاز الداخلي (داخل الاسطوانه)"

(A) Expansion



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Boyle's Law $\Rightarrow \boxed{P_1V_1 = P_2V_2}$

P_1 : internal pressure of gas

P_2 : Final pressure of gas = $P_{ext.}$

V_1 : initial volume of gas.

V_2 : Final volume of gas

Expansion	Compression
$V_2 > V_1$	$V_1 > V_2$
$P_1 > P_2$	$P_2 > P_1$
w: -ve	w: +ve

For the previous figure:

P_1 : P_{int} = 1400 KPa. $V_1 = 2L$

P_2 : P_{ext} = 600 KPa $V_2 = ?$

$$V_2 = \frac{P_1V_1}{P_2} = \frac{1400\,KPa \times 2L}{600\,L} = \boxed{4.67L}$$

$$w = -P_{ext} \cdot \Delta V = -600\,KPa * (4.67 - 2)L$$

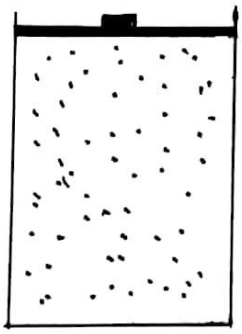
$$= -1602\ (KPa.L)$$

$$= -1602\ J$$

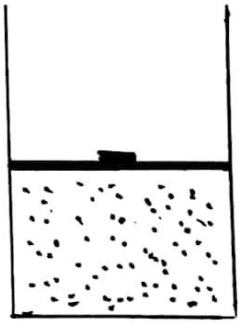
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(2) Compression:

$P_{ext} = 800\,KPa$



$6L = V_1$ ⟹ $V_2 = ?$



$P_{int} = 300\,KPa.$ $P_{int} = 800\,KPa.$

Boyle's Law: $P_1 V_1 = P_2 V_2$

$$300 \times 6L = 800 \times V_2$$

$$V_2 = 2.25L$$

$$W = -P_{ext}(V_2 - V_1)$$

$$= -800\,KPa * (2.25 - 6L)$$

$$= +3000\,J$$

Ex) Calculate work accompained with compression of a $3\,dm^3$ of a gas until its volume became $1\,dm^3$ given that its internal pressure = 1.3 atm

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$V_1 = 3L$

$V_2 = 1L.$

$P_1 = P_{int} = 1.3\ atm = 131.69\ KPa.$

$P_2 = P_{ext} = ?$

Boyle's Law $\Rightarrow$ $P_2 = P_{ext} = \frac{1.31.69\ KPa \times 3}{1L}$

$P_2 = P_{ext} = 395.07\ KPa.$

$W = -P_{ext}\ (V_2 - V_1)$

$= -395.07\ KPa * (1L - 3L)$

$= +790.14\ J$ (work done on system)

- Compression : ΔV: -ve (decrease)
  W : +ve.
  work done on system

- Expansion : ΔV: +ve (increase)
  W: -ve
  work done by system

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example:

A cylinder and piston is warmed up by an external flame. The contents (gas) expanded and pushed the piston upward against the external pressure.

If system absorbs 560 J of heat and the gas volume changes by 3.2L against external pressure of 200 KPa.

Calculate change in internal energy of the system.

$$\Delta E_{sys} = q + w$$

$$= (+560\,J) + (-200\,KPa \times +3.2\,L)$$

$$= -80\,J.$$

- Ideal gas $\Delta PE = 0$
- If ideal gas expands or compresses isothermally $(\Delta T = 0) \Rightarrow \Delta KE = 0$

$$\Delta E = \Delta PE + \Delta KE = q + w = 0$$

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Heat (q)

- Heat : form of energy which flows between two objects of different temperature.
  - Heat alway flow from the hotter object to the colder one.
- Heat unit is (J)
- Temperature : The degree of hotness or coldness of a body.
- Temperature reflects the average of Kinetic energy of molecules in a substance
- Temperature unit is (°C or K)

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How to calculate (q):

q = mass $\times$ specific heat $\times$ Δt

(g) $\left(\frac{J}{g \cdot °C}\right)$ °C

- specific heat : amount of heat needed to raise the temp. of one gram of substance by 1°C.
- Δt : $t_f - t_i$ (in °C)

q = heat capacity $\times$ Δt

$\left(\frac{J}{°C}\right)$ (°C)

- Heat capacity : amount of heat needed to raise temp. of substance by 1°C.

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ex) Calculate heat needed to raise the temp of 50g of water from 20°C to 50°C (sp.ht of water = $4.184 \frac{J}{g.°C}$)

q = mass × sp.ht × Δt

= $50g \times 4.184 \frac{J}{g.°C} \times (50-20)°C$

= $+6276\ J$

ex) Calculate the molar specific heat for a substance, if 16g of this substance (molar mass = 60g/mol) requires 260.8 J of heat to raise its temp. by 10°C.

q = moles × sp.ht × Δt

$+260.8J = \left(\frac{16}{60}\right) mol \times ? \times 10°C$

molar sp. ht = $97.8 \frac{J}{mol.°C}$ = $1.63 \frac{J}{g.°C}$

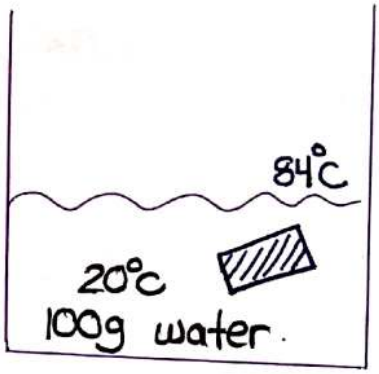
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Example: Suppose a 0.32 Kg rod of metal was inserted in 100g of water at 20°C. The temperature of the metal piece dropped down from 84°C to 33°C. If Sp. ht (H_2O) = 4.184 J/g.°C

Calculate the specific heat of this metal.

$$q_{sys} = -q_{surr.} \Rightarrow q_{water} = -q_{metal}$$

$$q_{water} = -q_{metal.}$$



Final temp for both = 33°C.

$$m \times \text{Sp.ht} \times \Delta t = -(m \times \text{sp.ht} \times \Delta t)$$

$$\left(100g \times 4.184 \frac{J}{g.°C} \times (33-20)°C\right) = \curvearrowright$$

$$-\left(320g \times ? \times (33-84°C)\right)$$

$$\Rightarrow 5439.2\ J = -(320g \times -51°C \times \text{sp.ht.})$$

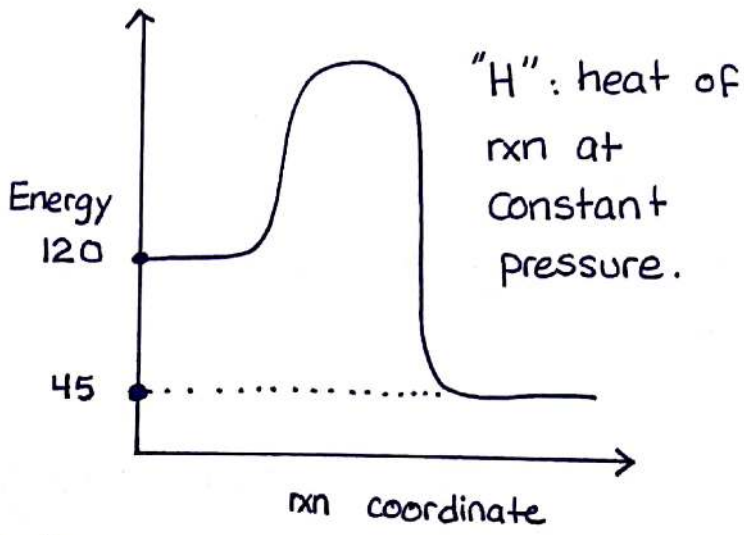
$$\text{Sp.ht} = 0.333 \frac{J}{g.°C}$$

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Lecture "11" online

Enthalpy (H)

- Heat of reaction is a state function



$\Delta H_{rxn.} = H_f - H_i$

$\Delta H_{rxn} = H_{products} - H_{reactants}$

$A + B \longrightarrow C.$

$\Delta H_{rxn} = 45 - 120$

$= -75$ KJ (exothermic rxn)

Ex) For the reaction $3A + 2B \longrightarrow C$

If 15g of A (F.wt = 72 g/mol) absorbs 53 KJ of heat. Calculate ΔH_{rxn}

$\frac{15g\ A}{72}$ = moles "A" = 0.208 mol A.

0.208 mol "A" $\Leftrightarrow$ +53 KJ

3mol "A" $\Leftrightarrow$?

$\Delta H = +764.4$ KJ

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$3A + 2B \longrightarrow C$ $\quad \Delta H = +764.4$ KJ

$6A + 4B \longrightarrow C$ $\quad \Delta H = (764.4 \times 2) = 1528.8$ KJ

$\frac{1}{2}C \longrightarrow B + \frac{3}{2}A$ $\quad \Delta H = (764.4 \times -\frac{1}{2}) = -382.2$ KJ

- Enthalpy (ΔH) is a quantitative property (extensive property).

ex) For a rxn $2A + B \rightarrow C$

If $\Delta H_{rxn} = -120$ KJ.

Calculate heat evolved when 20g of "A" (M.wt = 88g/mol) is reacted with sufficient amount of "B".

2 mol "A" $\Leftrightarrow$ -120 KJ.

$\left(\frac{20g}{88g/mol}\right) \Leftrightarrow$? (q)

$q = -13.6$ KJ.

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- ΔH tells if the rxn is:

  exothermic (ΔH : -ve) or

  endothermic (ΔH : +ve)

- ΔH_{rxn} is for a balanced chemical equation.

- ΔH_{rxn} depends on the coefficients of balanced equation.

- $\Delta H = \frac{q_{rxn}}{\text{moles}} \times \text{coefficient}$

  For any reactant or product in balanced equation.

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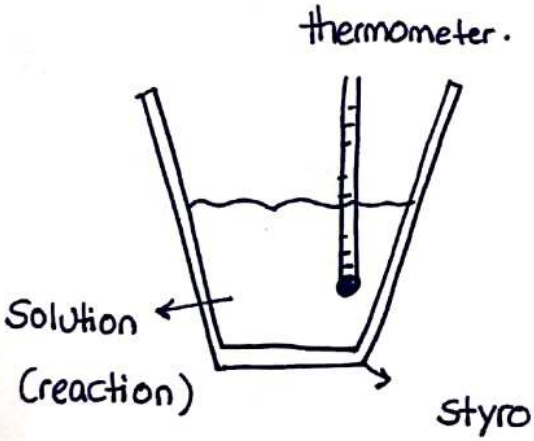
Calorimetery :

A method used to calculate enthalpy of a reaction usually by studying change in temperature of the surroundings.

$$q_{surr} = -q_{rxn.}$$

Calorimeter: device used to find the heat of reaction.

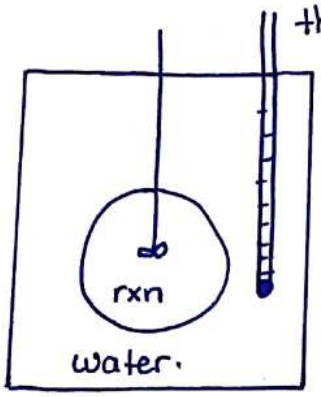
Calorimeters
- → Coffee-cup Calorimeter. (open system)
- → Bomb Calorimeter. (closed system).



⇒ Cup Calorimeter.

Surr : cup + air + water.

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Bomb Calorimeter.

⇒ Surr: Calorimeter + water.

" احياناً حتسب (q_{water}) لوحدها

واحياناً تعتبر مع صفن المسعر نفسه "

(1) Coffee - Cup Calorimeter:

⇒ open system ⇒ pressure is constant

⇒ "q" calculated is labeled as q_p

⇒ It is called solution calorimeter.

⇒ $q_{surr} = q_{air} + q_{cup} + q_{water}$.

⇒ To simplify calculations

water ≅ new solution.

$q_{water} = q_{new\ solution}$.

⇒ $q_{rxn} = -q_{surr}$ ⇒ $q_{p\ (rxn)}$

⇓

ΔH_{rxn}

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Lecture (12) : Thermodynamics

- Calculate work for chemical reactions:

$$\omega = -\Delta n_g \cdot R \cdot T$$

T: temp in Kelvin.

R: 8.314 $\frac{J}{mol.\ K}$

Δn_g : change in moles of gases.

ex) Calculate work for the following rxns:

$$NaOH_{(aq)} + H_2SO_{4\,(aq)} \longrightarrow Na_2SO_{4\,(aq)} + 2H_2O_{(\ell)}$$

$$\Delta n_g = 0 - 0 = 0.$$

$$\omega = -\Delta n_g \cdot R \cdot T = 0.$$

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ex) For the rxn:

$3H_{2\,(g)} + N_{2\,(g)} \longrightarrow 2NH_{3\,(g)}$ at 25°C

$\Delta n_g = (2\ mol) - (4\ mol) = -2\ mol.$

$w = -\Delta n_g \cdot R \cdot T$

$= -(-2\ mol) \times 8.314 \frac{J}{mol.K} \times 298\ K$

$= 4955.14\ J$

For the rxn:

$C_6H_{6\,(\ell)} + \frac{15}{2}O_{2\,(g)} \rightarrow 6CO_{2\,(g)} + 3H_2O_{(g)}$

$\Delta n_g = (3+6) - (7.5+0) = +1.5\ mol$

$w = -(1.5\ mol) \times 8.314 \frac{J}{mol.K} \times 298\ K$

$= -3716.4\ J.$

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Δn_g	w	
+	−	done by system (expansion)
−	+	done on system (compression)

Calorimetery :

1) Open system : Cup - Calorimeter.

2) Closed system : Bomb Calorimeter.

In both calorimeters you study and determine q_{surr} ⇒ which equals $-q_{rxn.}$ ⇒ $q_{surr} = -q_{surr}$

q_{surr} → mass × sp.ht × Δt

q_{surr} → heat cap. × Δt

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(1) Cup - Calorimeter. (Solution calorimeter)

Soln (1) + soln (2) ⟶ new soln.

↓

(surrounding)

$$q_{surr} = q_{air} + q_{cup} + q_{new\ soln.}$$

q_{cup} ← أحياناً يُحسب

$$q_{new\ soln} = \text{mass (soln)} \times \text{sp.ht (soln)} \times \Delta t$$

mass (soln) (V x d)

sp.ht (soln)

Δt فرق الحرارة قبل وبعد التفاعل (°C)

$$q_{p\ (rxn)} = -q_{new\ soln.}$$

$q_{p\ (rxn)}$ ⇒ كمية معينة من المادة تم استخدامها

$$\Delta H \equiv q_{p\ (rxn)}$$

$$\Delta H = \frac{q_{(rxn)}}{\text{moles}} \times \text{coefficient}$$

moles ↘ لأي ماده في التفاعل (للعامل المحدد)

coefficient ↙

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Example :

Calculate ΔH_{rxn} for the following reaction given that 50mL (0.1M) of NaOH were mixed with 25mL (0.1M) of H_2SO_4 in a cup calorimeter and the temp. raised from 22°C to 35°C.

Useful data: Sp.ht of Na_2SO_4 Soln = $4.08 \frac{J}{g.°C}$

density of Na_2SO_4 Soln = $1.07 \frac{g}{mL}$.

$$2NaOH_{(aq)} + H_2SO_4{}_{(aq)} \longrightarrow \underbrace{Na_2SO_4{}_{(aq)} + H_2O_{(l)}}$$

50mL (0.1M) 22°C	25mL (0.1M) 22°C.	75mL 35°C.

$$q_{surr} = q_{air} + q_{cup} + q_{new\ soln.}$$

q_{air} (neglect); q_{cup} neglected sometimes; $q_{new\ soln.}$ ✓

$$q_{surr} = (m \times \text{Sp.ht} \times \Delta t)_{Na_2SO_4\ soln.}$$

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$$q_{surr} = (V \times d) \times 4.08 \frac{J}{g \cdot ^\circ C} \times (35 - 22)^\circ C$$

$$= (75 \times 1.07)\,g \times 4.08 \frac{J}{g \cdot ^\circ C} \times 13^\circ C.$$

$$= 4256.46\ J \Rightarrow (+ve)$$
$$4256.46\ J$$

$$q_{rxn} = -4256.46\ J.$$

5×10^{-3} mol NaOH

2.5×10^{-3} mol H_2SO_4.

For the rxn: $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

$$\Delta H_{rxn} = \frac{q_{rxn}}{\text{moles}} \times \text{coefficient}$$

$$= \frac{-4256.46\ J}{5 \times 10^{-3}\ mol} \times 2$$

$$= -1702.58\ KJ$$

$$\Delta H\ (KJ/mol\ NaOH) = -851.29\ KJ.$$

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(2) Bomb Calorimeter

System not at constant pressure.

System is at constant volume.

$$q_{surr} = q_{water} + q_{calorimeter}.$$

$$q_{v\,(rxn)} = -q_{surr}$$

$$\Delta V = 0 \Rightarrow w = -P_{ext.}\,\Delta V = 0.$$

$$\Delta E_{rxn} = q_{v(rxn)} + w_{(rxn)}$$

$$\Delta E_{rxn} = \frac{q_{v\,(rxn)}}{\text{moles}} \times \text{coefficient.}$$

$q_{v\,(rxn)}$ ⇒ حرارة التفاعل عند تفاعل كمية معينه من المادة ، كلما التفاعل الكيميائي الموزون فيه مولات محدده (نسبة وتناسب)

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Lecture "13"

(2) Bomb Calorimeter

System not at constant pressure.

System is at constant volume.

$$q_{surr} = q_{water} + q_{calorimeter}.$$

$$q_{v\,(rxn)} = -q_{surr}$$

$$\Delta V = 0 \Rightarrow w = -P_{ext.}\,\Delta V = 0.$$

$$\Delta E_{rxn} = q_{v(rxn)} + w_{(rxn)}$$

$$\Delta E_{rxn} = \frac{q_{v\,(rxn)}}{\text{moles}} \times \text{coefficient.}$$

$q_{v\,(rxn)}$ ⇒ حرارة التفاعل عند تفاعل كمية معينه من المادة ، كل التفاعل الكيميائي الموزون فيه مولات محدده (نسبة وتناسب)

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Example: For the rxn:

$$C_3H_7OH_{(\ell)} + 5O_{2\,(g)} \longrightarrow 3CO_{2\,(g)} + 4H_2O_{(g)}$$

When 1.325 g of C_3H_7OH (60 g/mol) were allowed to react with suffecient amount of oxygen in a bomb calorimeter the temperature of 1 Kg of water in the calorimeter raised from 22°C to 38°C. If heat capacity of calorimeter = 42 $\frac{KJ}{°C}$ and sp.ht of water = 4.184 $\frac{J}{g \cdot °C}$

Calculate (a) ΔE_{rxn} (in KJ)

(b) ΔE_{rxn} (in $\frac{KJ}{mol\ H_2O}$)

(c) ΔH_{rxn} (in KJ)

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$$q_{surr} = q_{water} + q_{cal.}$$

$$= (m \times \text{sp.ht} \times \Delta t) + (\text{heat. cap.} \times \Delta t)$$

$$= \left(1000\,g \times 4.184 \frac{J}{g.^\circ C} \times 16^\circ C\right) + \left(42 \frac{KJ}{^\circ C} \times 16^\circ C\right)$$

$$= 66944\ J + 672\ KJ$$

$$= 738.94\ KJ \quad (+ve)$$

$$q_{v\,(rxn)} = -738.94\ KJ \quad (-ve)$$

$$\Delta E = \left(\frac{q_v}{\text{moles}} \times \text{coeff.}\right) + \text{work} \rightsquigarrow \text{zero.}$$

$$\Delta E = \frac{-738.94\ KJ}{(1.325g / 60\,g/mol)} \times 1.$$

$$= -33504.9\ KJ = -33504.9\ \frac{KJ}{mol\ C_3H_7OH}$$

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(b) ΔE (KJ/mol H_2O)

$\Delta E \Rightarrow -33504.9$ KJ $\Leftrightarrow$ 4 mol H_2O

$? \Leftrightarrow$ 1 mol H_2O

$\Delta E = -8376$ KJ.

(C) ΔH_{rxn} (in KJ)

$\Delta E = \Delta H + -\Delta n_g . R T$

$(4+3) - 5 = +2$

$-33504.9 = \Delta H + -(2 \times 8.314 \frac{J}{mol.K} \times 295 K)$

$\Delta H = \Delta E + \Delta n_g RT$

$= -33504.9 \text{ KJ} + 4905.26 \text{ J}$

$= -33499.99 = -33500$ KJ

exothermic rxn

ΔH: -ve

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$$\Delta E = \Delta H + -\Delta n_g . R . T \quad \text{(for rxn)}$$

$$\Delta H = \Delta E + \Delta n_g . R . T \quad \text{(for rxn)}$$

T: temp of rxn in Kelvin.

$$\Delta E^\circ = \Delta H^\circ - \Delta n_g \times R \times 298\,K$$

$^\circ$: means standard conditions.

(STP)

standard temperature (25°c)

and standard pressure (1 atm)

For the rxn: $2A_{(aq)} + 3B_{(g)} \rightarrow C_{(g)} + \frac{1}{2} D_{(g)}$

$$\Delta H = \Delta E + -1.5\,RT = \Delta E - 1.5\,RT$$

$$\Delta E = \Delta H + 1.5\,RT$$

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Rate Law : قانون السرعة

For the rxn: $2A + B \rightarrow C$

The rate law expression is:

$$\text{Rate} = K[A]^n[B]^m$$

- K: rate constant
- K: is temperature dependent.

  "Usually as temp increase ,"K" increases."
- [A] ,[B] are initial concentrations of the reactants A, and B.
- "conc. of products does not appear in the rate law expression.
- "n" order of reaction with respect to "A"
- m: order of reaction with respect to "B".
- The overall order of reaction = n+m.

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How to find the orders of reaction for the rate Law expression.

- Given the reaction:

$$5\,Fe^{2+} + MnO_4^- + 8H^+ \longrightarrow 5\,Fe^{3+} + Mn^{2+} + 4H_2O$$

Trial	$[Fe^{2+}]$ (M)	$[MnO_4^-]_o$ (M)	$[H^+]_o$ (M)	Rate of rxn (M/s)
1.	0.2	0.3	0.2	4×10^{-4}
2.	0.4	0.3	0.2	3.2×10^{-3}
3.	0.2	0.9	0.2	3.6×10^{-3}
4.	0.4	0.3	0.6	3.2×10^{-3}

Write the rate law expression for the above rxn

$$\text{Rate} = K\,[Fe^{2+}]^n\,[MnO_4^-]^m\,[H^+]^p$$

- To find "n" use trial "1" and "2"

$$\frac{\text{Rate (2)}}{\text{Rate (1)}} = \frac{3.2 \times 10^{-3}}{4 \times 10^{-4}} = \frac{K\,[0.4]^n\,[0.3]^m\,[0.2]^p}{K\,[0.2]^n\,[0.3]^m\,[0.2]^p}$$

$$8 = \left(\frac{0.4}{0.2}\right)^n$$

$$8 = 2^n$$

$$n = 3$$

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- To find order of "MnO_4^-", use trials 1,3.

$$\left(\frac{0.9}{0.3}\right)^m = \frac{3.6\times10^{-2}}{3.2\times10^{-3}}$$

$$(3)^m = 9 \quad \Rightarrow \boxed{m=2}$$

- To find order of "H^+", (p), use trials 2,4.

$$\left(\frac{0.6}{0.2}\right)^p = \frac{3.2\times10^{-3}}{3.2\times10^{-3}} = 1$$

$$3^p = 1 \quad \Rightarrow \boxed{p = \text{zero}.}$$

The rate law expression becomes:

$$\text{Rate} = K\,[Fe^{3+}]^3\,[MnO_4^-]^2\,[H^+]^0$$

$$\text{Rate} = K\,[Fe^{3+}]^3\,[MnO_4^-]^2$$

The overall order of reaction is "5"
(Fifth order rxn)

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- n, m : order could be +ve, -ve integers, zero, any fraction.

- n, m are determined experimentally
- n, m ≠ coefficients of "A" and "B" in the balanced equation.

- Note: Unit of "K" is related to the overall order of the reaction.

$$\text{Unit of "K"} = M^{1-x} \cdot \text{time}^{-1}$$

- where M: is molarity = $mol.L^{-1}$

x: overall order of the rxn.

time: s^{-1}, hr^{-1}, min^{-1}

example: For a 4th order rxn:

$$\text{Unit of } K = M^{-3} \cdot s^{-1}$$

$$= \boxed{mol^{-3} \cdot L^{3} \cdot s^{-1}}$$

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To find value of "K"

Use any of the trials in the previous table.

ex. trial (1)

$$\text{Rate} = K\,[Fe^{3+}]^3\,[MnO_4^-]^2$$

$$4\times10^{-4}\,\frac{M}{S} = K\cdot(0.2M)^3\cdot(0.30)^2$$

$$K = 0.56\times10^{-4}\cdot S^{-1}\cdot M^{-4}$$

القيمه والوحدة مهمة .

Note: When concentration of one of the reactants, changes, and the rate is kept constant (unchanged)

Then the rxn is zeroth order with respect to this reactant.

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- Given the reaction :

$$A + B \longrightarrow C \qquad K = 4.9 \times 10^{-3}\ M^{-2} \cdot s^{-1}$$

When the concentration of "B" was doubled the rate of reaction was also doubled.

Calculate the rate of reaction when the initial concs of "A" and "B" are 0.3M and 0.2M respectively.

Notice that :

- The reaction is third order overall (from unit of K)

- The rxn is first order with respect to "B"

$$\left(\begin{array}{c}\text{Conc. change} \\ \text{of "B"}\end{array}\right)^{m} = \text{rate change}$$

$$(2)^{m} = 2 \Rightarrow m = 1$$

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- The rate law is:

$$\text{Rate} = K[A]^n[B]^m$$

$$\text{Rate} = K[A]^n[B]^1$$

So $n = 2$, because rxn is third order.

$$\text{Rate} = 4.9\times10^{-3} \times (0.3M)^2 \cdot (0.2M)^1$$

$$= \boxed{8.82\times10^{-5}\ M/s.}$$

ex) What is the order of reaction

$A \longrightarrow B$ given that,

When conc. of "A" is tripled, the rate increased by 27 times.

$$(A)^x = \text{rate change.}$$

$$(3)^x = 27$$

$$\boxed{x = 3}$$ (third order rxn)

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Integrated rate Law:

The integrated rate law describes how concentration of a reactant changes with time.

هذا القانون يمكننا من حساب الكمية المتبقية من مادة متفاعلة بعد زمن معين.

- The integrated law expression for a reaction depends on the overall order of reaction.

The reaction studied is $A \longrightarrow B$

$[A]_0$: initial conc. of A.

$[A]_t$: conc. of "A" remained after "t" time.

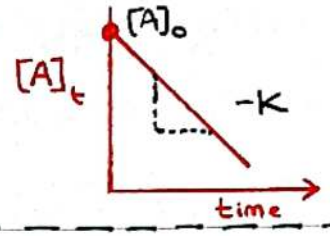
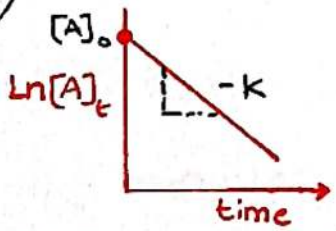
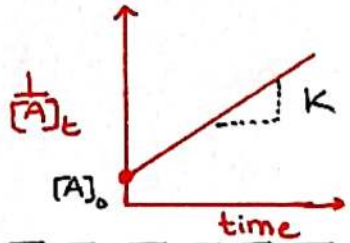
$$[A]_t < [A]_0$$

$$[A]_t = x\,[A]_0 \Rightarrow x \leq 1$$

"x" : fraction of reactant "A" remained after "t" time. (ex. $x = \frac{1}{3}, \frac{1}{2}$, 20%, 30%)

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Integrated rate Law for: $A \longrightarrow B$

order	Rate Law	Integrated rate law	Half-life ($t_{1/2}$)
Zeroth	Rate = K (K: $M.s^{-1}$)	$[A]_t - [A]_o = -Kt$ $[A]_t = [A]_o - Kt$ 	$t_{1/2} = \frac{[A]_o}{2K}$
First	Rate = K [A] K: s^{-1}	$Ln\left(\frac{[A]}{[A]_o}\right) = -Kt$ 	$t_{1/2} = \frac{Ln(2)}{K}$
Second	Rate = K $[A]^2$ (K: $M^{-1}.s^{-1}$)	$\frac{1}{[A]_t} - \frac{1}{[A]_o} = Kt$ 	$t_{1/2} = \frac{1}{K[A]_o}$

$t_{1/2}$: Half life of rxn: it is time needed to consume 50% of the initial concentration of reactant "A"

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$[A]_c$: consumed concentration of (A) after "t" time.

$$[A]_c = [A]_o - [A]_t$$

$[A]_o$: التركيز الابتدائي للمتفاعل (A)

$[A]_t$: التركيز المتبقي من (A) في وعاء التفاعل بعد زمن معين

- Example: For a zeroth order rxn:

If concentration of reactant (A) drops down from 0.62M to 0.14M within 12.5 minutes. Find half life of reaction.

$$t_{1/2} = \frac{[A]_o}{2K}$$

but "K" rate constant is not mentioned, so it has to be calculated.

For zeroth order:

$$[A]_t - [A]_o = -Kt$$

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$$[A]_t - [A]_o = -kt$$

$$(0.14M - 0.62M) = -K \times (12.5\ min)$$

$$K = \frac{0.48\ M}{12.5\ min} = 3.84 \times 10^{-2}\ M/min.$$

$$t_{1/2} = \frac{[A]_o}{2K} = \frac{0.62M}{2 \times 3.84 \times 10^{-2}\ M/min}$$

$$t_{1/2} = 8.07\ minutes = \underline{484.4\ s}$$

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For the rxn: $A \rightarrow B$

If $K = 0.005\ s^{-1}$

Answer the following :

a. Find first, second, and third half lifes of the rxn.

$$t_{1/2} = \frac{Ln\ 2}{K} = \frac{0.693}{0.005\ s^{-1}} = 138.6\ s.$$

For 1st order rxns :

First $t_{1/2}$ = Second $t_{1/2}$ = 3rd $t_{1/2}$ = n^{th} $t_{1/2}$.

↓ ↓

138.6 s. 138.6 s.

b. At what time is the rxn 46% complete?

46% complete means → $\frac{46}{100}$ $[A]_o$ is reacted (consumed)

→ $\frac{54}{100}$ $[A]_o$ is remained.

دائماً نخوض التركيز المتبقي في قانون السرعه (وليس المستهلك)

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- 1st order Law:

$$\ln \frac{[A]_t}{[A]_0} = -kt$$

$$\ln \left(\frac{0.54\ A_0}{A_0}\right) = -0.005\ s^{-1} \times time$$

$$time = \frac{\ln (0.54)}{-0.005\ s^{-1}} = 123.23\ s$$

- أي أنه بعد ١٢٣ ثانيه سيستهلك 46% من التركيز الابتدائي للمادة المتفاعلة ، وسيتبقى فقط 54% منها في وعاء التفاعل .

(c) What fraction (percentage) of the initial concentration is consumed after 1.4 minutes.

$$[A]_t = x\ [A]_0$$

$$\ln \left(\frac{[A]_t}{[A]_0}\right) = -kt \Rightarrow \ln \left(\frac{x\ A_0}{A_0}\right) = -kt$$

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$$\ln(X) = -0.005\,s^{-1} \times \text{time}.$$

$$\ln(X) = -0.005\,s^{-1} \times (1.4\,\text{min} \times 60)\,s$$

$$\ln(X) = -0.42.$$

To get (X), on your calculator press

Shift → Ln → -0.42 to get the answer.

$$X = 0.65 = 65\% \text{ remained.}$$

The fraction consumed $= 1 - 0.65$

$$= 0.35$$

$$= 35\%$$

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For the second order reaction:

$$2HI \longrightarrow H_2 + I_2$$

If initial concentration of HI = 0.6M and the first half life of the reaction = 0.98 hr. Answer the following:

(a) Find concentration of (HI) consumed after half an hour.

$$\frac{1}{[A]_t} - \frac{1}{[A]_o} = kt$$

K?

You can find "K" from $t_{1/2}$:

$$t_{1/2} = \frac{1}{K[HI]_o} \Rightarrow 0.98hr = \frac{1}{K*(0.6M)}$$

$$K = 1.7\ M^{-1}.hr^{-1}$$

$$\frac{1}{[HI]_t} - \frac{1}{0.6} = 1.7\ M^{-1}.hr^{-1} * 0.5hr.$$

$$[HI]_t = \underline{0.4M} \Rightarrow$$

this is conc. remained

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$$[HI]_{consumed} = [HI]_0 - [HI]_{remained}.$$

$$= 0.6M - 0.4M$$

$$= 0.2M$$ concentration consumed.

(b) Find the fourth half life of the rxn:

$$[HI]_0 \xrightarrow[(t_{1/2})]{\text{First}} \frac{1}{2}[HI]_0 \xrightarrow[(t_{1/2})]{\text{Second}} \frac{1}{4}[HI]_0 \xrightarrow[(t_{1/2})]{\text{third}} \frac{1}{8}[HI]_0$$

0.98 (0.98x2) (0.98x4)

$$\downarrow \text{fourth } (t_{1/2})$$

$$\frac{1}{16}[HI]_0 \quad (0.98 \times 8)$$

$$\frac{1}{[HI]_t} - \frac{1}{[HI]_0} = kt.$$

$$\frac{1}{\frac{1}{16}[HI]_0} - \frac{1}{[HI]_0} = kt.$$

$$\frac{1}{0.0375M} - \frac{1}{0.6} = 1.7\ M^{-1}.hr^{-1} \times t$$

(fourth $t_{1/2}$)

$$t = 7.84\ hr = 470.4\ min.$$

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(c) Find percentage of HI remained after 130 minutes (notice: K is in hours).

$t = 130\text{min} / 60 = 2.167 \text{ hrs.}$

$$\frac{1}{[HI]_t} - \frac{1}{[HI]_0} = Kt$$

$$\frac{1}{x\,[HI]_0} - \frac{1}{[HI]_0} = Kt$$

$$\frac{1}{0.6\,x} - \frac{1}{0.6M} = 1.7\,M^{-1}\,hr^{-1} \times 2.167\,hr.$$

$$x = \frac{1}{0.6 \times 3.68} = 0.45$$

$= \underline{45\%}$ is remained.

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About half lifes of reactions:

Half life.	Zeroth order	First order	Second order.
First	$\frac{[A]_0}{2K}$	$\frac{Ln\ (2)}{K}$	$\frac{1}{K[A]_0}$
	$\downarrow \times \frac{1}{2}$	$\downarrow \times 1$	$\downarrow \times 2$
Second	$\frac{[A]_0}{4K}$	$\frac{Ln\ (2)}{K}$	$\frac{2}{K[A]_0}$
	$\downarrow \times \frac{1}{2}$	$\downarrow \times 1$	$\downarrow \times 2$
third	$\frac{[A]_0}{8K}$	$\frac{Ln\ (2)}{K}$	$\frac{4}{K[A]_0}$
	$\downarrow \times \frac{1}{2}$	$\downarrow \times 1$	$\downarrow \times 2$
fourth	$\frac{[A]_0}{16K}$	$\frac{Ln\ (2)}{K}$	$\frac{8}{K[A]_0}$

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Activation Energy (E_a)

Activation energy is minimum energy required to start up reaction.

It is energy required to break up bonds in reactants and form new bonds in products.

Ea is always a positive value. $\left(\frac{J}{mol.}\right)$ or $\left(\frac{KJ}{mol}\right)$

- How to calculate E_a for a reaction.

$$Ln\left(\frac{K_2}{K_1}\right) = \frac{-Ea}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

- T_1: lower temperature in Kelvin.
- T_2: higher temperature in Kelvin.
- K_1: rate constant at T_1.
- K_2: rate constant at T_2.
- R: Universal gas constant = 8.314 $\frac{J}{mol.K}$

$T_2 > T_1$

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Arrhenius Equation

$$\ln K = \ln A - \frac{Ea}{RT}$$

- K: rate constant at (T)
- A: Arrhenius constant.
- Ea: Activation energy in $\left(\frac{J}{mol}\right)$
- R: 8.314 $\frac{J}{mol.K}$

Ex) For the reaction: $A + B \rightarrow C$

If K = 0.035 at 25°C and K = 0.21 hr^{-1} at 42°C. Find the activation energy of reaction in $\left(\frac{KJ}{mol}\right)$

$T_1 = 25°C = 298\,K \Rightarrow K_1 = 0.035\ hr^{-1}$

$T_2 = 42°C = 315\,K \Rightarrow K_2 = 0.21\ hr^{-1}$

$$\ln\left(\frac{0.21}{0.035}\right) = \frac{-Ea}{8.314} \times \left(\frac{1}{315} - \frac{1}{298}\right)$$

$$Ea = 82256.0\ \frac{J}{mol} = +82.25\ \frac{KJ}{mol}.$$

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Ex) For a reaction: $A + B \rightarrow C$.

If rate constant $K = 0.035\ M/s$ at 25°C.

At what temperature is the rate constant tripled, given that activation energy = 62 kJ/mol.

$$\ln\left(\frac{K_2}{K_1}\right) = \frac{-Ea}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

$T_1 = 25°C = 298\ K \Rightarrow K_1 = 0.035\ M/s$

$T_2 = ?$

$\Rightarrow K_2 = 3K_1$

$= 0.0117\ M/s$

$$\ln\left(\frac{3K_1}{K_1}\right) = \frac{-62000\ J/mol}{8.314\ J/mol.K} \times \left(\frac{1}{T_2} - \frac{1}{298}\right)$$

$$\ln(3) = -7457.3 \times \left(\frac{1}{T_2} - 3.36 \times 10^{-3}\right)$$

$T_2 = 311.3\ K$

$T_2 = 38.3°C.$ ✓

$A = 2.58 \times 10^9$ (Arrhenius constant)

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Catalysts: الحوامل المحفزة للتفاعل

Catalysts are chemical substances added to reaction to speed up reaction.

"To increase rate of reaction"

- The catalyst speeds up reaction by decreasing the activation energy of reaction without itself being consumed.
- Catalysts lower Ea of the reaction.

Types of Catalysts:

(1) Homogeneous Catalysts: these have the same phase (physical state) as reactants.

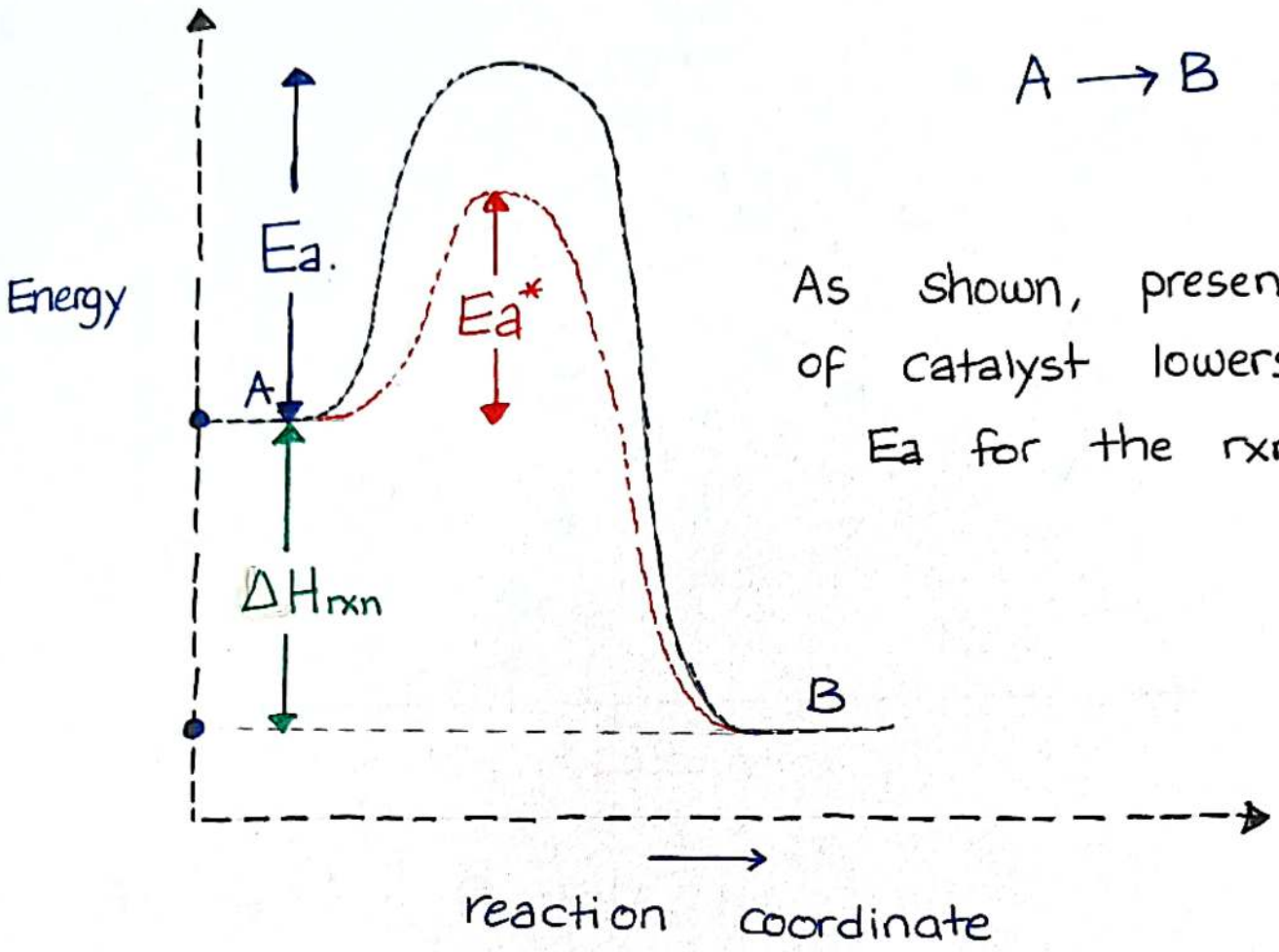
(2) Heterogeneous Catalysts: these have a different phase from that for reactants.

$$A_{(\ell)} + B_{(\ell)} \xrightarrow{\text{Catalyst}_{(\ell)}} \text{products} \quad \text{(Homogeneous)}$$

$$A_{(\ell)} + B_{(\ell)} \xrightarrow{\text{Catalyst}_{(s)}} \text{products} \quad \text{(Heterogeneous)}$$

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Potential Energy Diagram



As shown, presence of catalyst lowers Ea for the rxn.

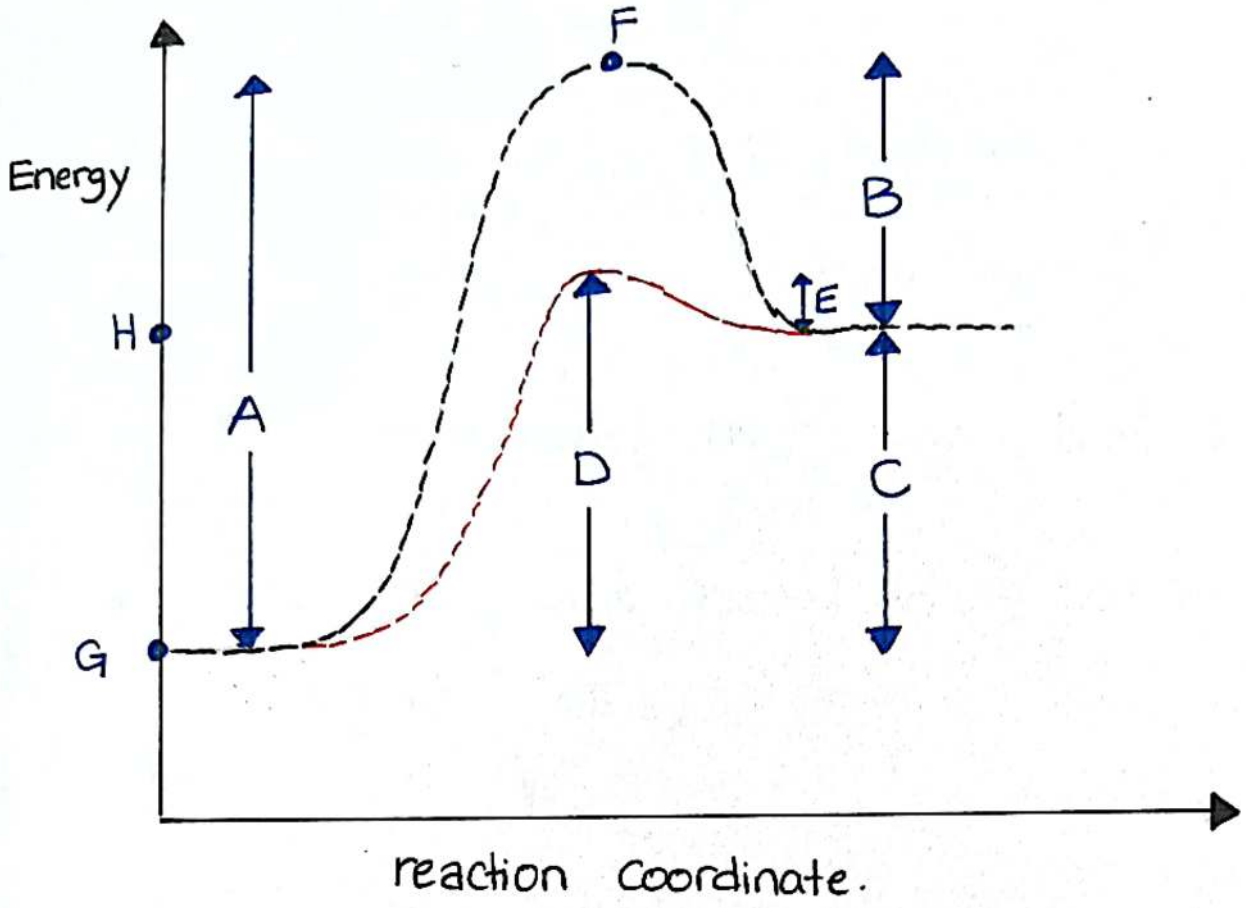
Ea: activation energy of forward reaction.

Ea*: activation energy of forward reaction in presence of catalyst.

ΔH: in this rxn is (-ve) (exothermic reaction.)

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For the following energy diagram:



- A: Ea for forward reaction
- B: Ea for reverse reaction
- C: ΔH reaction (here it is endothermic)
- D: Ea for forward rxn in presence of catalyst.
- E: Ea for reverse rxn in presence of catalyst.
- F: Activated complex (المعقد المنشط)
- G: Energy of reactants (forward rxn)
- H: Energy of products (forward rxn).

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Important notes:

- Fast reactions have high rate values.
- Fast reactions have high rate constants (K)
- Fast reactions have low half lifes ($t_{1/2}$)
- Fast reactions have low activation energies (Ea)

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